# The UC models of `categorical-crypto`

Setups, the morphisms between them, and the major theorems

Branch `uc-setup-landscape`, 10 October 2026

This note is a self-contained mathematical account of the models of universally composable (UC) security — *setups* — that the library formalises, of the structure-preserving maps between them, and of the theorems proved about them. It is written for a reader who knows category theory and cryptography and does not read Agda: it says nothing about how the formalisation is organised.

**Names and links.** Every setup, setup morphism and major theorem has one mathematical name, fixed in Section 2 and Section 5 and used throughout. Every formal statement ends with a list (Agda: links) to the Agda identifiers that define or prove it on branch `uc-setup-landscape`; the link text is the identifier, the target its defining line.

**Status.** FORMAL marks what is proved in the tree. DESIGN marks what comes from a design document (`docs/uc-setup-*.md`, `docs/uc-robust-compilation.md`): it has not been type-checked and the argument given is informal. INSTANCE marks a statement that is not stated separately in the tree but is the instantiation of a proved general theorem at proved hypotheses, all of which are linked. A statement with neither tag nor link is a definition or notation made for this exposition.

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# 1 The abstract framework

## 1.1 UC setups

**Definition 1** (UC setup). A *UC setup* $\mathbb{S} = (\mathcal{C}, \mathcal{I}, \mathcal{M}, \mathcal{E})$ consists of a category $\mathcal{C}$ of computations; a monoidal category $(\mathcal{I}, \otimes, I)$ of interfaces, used as *grades*; an $\mathcal{I}$-graded monad $\mathcal{M}$ on $\mathcal{C}$, that is, a lax monoidal functor $\mathcal{M} : \mathcal{I} \to ([\mathcal{C}, \mathcal{C}], \circ, \mathrm{Id})$; and a presheaf of setoids $\mathcal{E} : \mathcal{C}^{\mathrm{op}} \to \mathsf{Setoid}$, the *tests*. (Agda: `UCSetup`)

We write $T_X := \mathcal{M}(X)$, $\eta : \mathrm{Id} \Rightarrow T_I$ and $\mu_{X,Y} : T_X T_Y \Rightarrow T_{X \otimes Y}$ for the lax structure, and $\mathrm{sub}_c := \mathcal{M}(c) : T_X \Rightarrow T_{X'}$ for $c : X \to X'$ in $\mathcal{I}$. The Kleisli extension of $h : B \to T_Y C$ at grade $X$ is $\mathrm{ext}_X\, h := \mu_{X,Y} \circ T_X h$, and for $f : A \to T_X B$ the *serial composite* is $h \bullet f := \mathrm{ext}_X\, h \circ f : A \to T_{X \otimes Y} C$ (Agda: `_•_`). Graded monads are those of [1].

The cryptographic reading is fixed once and for all.

- A *protocol* (or functionality) from $A$ to $B$ with *adversary interface* $X$ is a morphism $f : A \to T_X B$ of $\mathcal{C}$.
- An *adversary* is a morphism $a : X \to X'$ of $\mathcal{I}$; it attacks $f$ by regrading, $\mathrm{sub}_a \circ f$. A *simulator* is the same kind of object, acting on the ideal protocol.
- A *test* on $A$ is an element $e \in \mathcal{E}(A)$; a morphism $h : A' \to A$ acts on tests by restriction, $e \cdot h := \mathcal{E}(h)(e)$.
- For $W \in \mathcal{I}$ (an *ancilla*), the *prefix context* of $f$ is $\mu_{W,X} \circ T_W f : T_W A \to T_{W \otimes X} B$, and $\mathrm{run}_W(f)(e) := e \cdot (\mu_{W,X} \circ T_W f)$ runs $f$ inside the environment $e \in \mathcal{E}(T_{W \otimes X} B)$ (Agda: `prefix`, `run`).

## 1.2 Two equivalences

**Definition 2** (observational kernel). For parallel $f, g : A \to B$ in $\mathcal{C}$, $f \approx_{\mathcal{E}} g$ iff $\mathcal{E}(f) = \mathcal{E}(g)$, i.e. $e \cdot f \approx e \cdot g$ in $\mathcal{E}(A)$ for every $e \in \mathcal{E}(B)$. It is the kernel congruence of the functor $\mathcal{E}$, hence a congruence on $\mathcal{C}$ containing its hom-equality. (Agda: `_~_`)

**Definition 3** (U-kernel). For $f, g : A \to T_X B$, $f \approx^{\mathrm{U}} g$ iff $\mu_{W,X} \circ T_W f \approx_{\mathcal{E}} \mu_{W,X} \circ T_W g$ for every $W \in \mathcal{I}$: the two protocols agree in every environment at every ancilla. (Agda: `_≈ᵁ_`)

The name: $\approx^{\mathrm{U}}$ is the kernel of the enriched forgetful functor that sends a graded morphism to the family of its prefix contexts; the underlying locally graded structure is that of McDermott and Uustalu (MPC 2022).

**Proposition 4** (congruence). $\approx^{\mathrm{U}}$ is an equivalence relation that contains the hom-equality and is contained in $\approx_{\mathcal{E}}$. It is a congruence for regrading ($f \approx^{\mathrm{U}} f' \Rightarrow \mathrm{sub}_c \circ f \approx^{\mathrm{U}} \mathrm{sub}_c \circ f'$), for Kleisli extension ($f \approx^{\mathrm{U}} f' \Rightarrow \mathrm{ext}_X\, h \circ f \approx^{\mathrm{U}} \mathrm{ext}_X\, h \circ f'$) and for serial composition in the other argument ($h \approx^{\mathrm{U}} h' \Rightarrow h \bullet f \approx^{\mathrm{U}} h' \bullet f$). (Agda: `≈C⇒≈ᵁ`, `≈ᵁ⇒≈ℰ`, `sub-cong`, `ext-cong`, `•-cong-arg`)

**Definition 5** (grade stability). $\mathbb{S}$ is *grade-stable* if every $T_Y$ respects $\approx_{\mathcal{E}}$: $h \approx_{\mathcal{E}} h' \Rightarrow T_Y h \approx_{\mathcal{E}} T_Y h'$. Equivalently, the kernel of $\mathcal{E}$ is a congruence of $\mathcal{M}$-modules. (Agda: `GradeStable`)

**Proposition 6** (bridges from $\approx_{\mathcal{E}}$ to $\approx^{\mathrm{U}}$). (i) In a grade-stable setup, $f \approx_{\mathcal{E}} g \Rightarrow f \approx^{\mathrm{U}} g$ for graded $f, g$; with Proposition 4 the two relations then coincide on graded morphisms. (ii) If $\mathcal{E}$ is faithful, $\mathbb{S}$ is grade-stable. (iii) If the kernel is generated by a conjugation-closed relation that is detected at every $T_V$, and every $\mu_{X,Y}$ has a retraction, $\mathbb{S}$ is grade-stable. (Agda: `bridge`, `faithful⇒grade-stable`, `GradeStableFromTests`)

Grade stability is used by nothing in the metatheory of Section 1.3. It matters only when a fact proved at the bare kernel $\approx_{\mathcal{E}}$ has to be imported into $\approx^{\mathrm{U}}$, and when a morphism is asked to carry emulation (Theorem 10). The main model of the library is *not* grade-stable (Proposition 18), which shapes much of what follows.

## 1.3 Emulation and the metatheory

**Definition 7** (emulation). For $f : A \to T_X B$ and $g : A \to T_Y B$, $f \leq_{\mathrm{UC}} g$ iff for every adversary $a : X \to X'$ there is a simulator $s : Y \to X'$ with $\mathrm{sub}_a \circ f \approx^{\mathrm{U}} \mathrm{sub}_s \circ g$. (Agda: `_≤UC_`)

**Theorem 8** (UC metatheory). In every UC setup, with no hypotheses:

1. $\leq_{\mathrm{UC}}$ is a preorder, invariant under the hom-equality, and $f \approx^{\mathrm{U}} g \Rightarrow f \leq_{\mathrm{UC}} g$.
2. *Dummy-adversary completeness.* $f \leq_{\mathrm{UC}} g$ iff there is $s_0 : Y \to X$ with $f \approx^{\mathrm{U}} \mathrm{sub}_{s_0} \circ g$; the simulator for an adversary $a$ is then $a \circ s_0$.
3. *Composition.* If $f \leq_{\mathrm{UC}} g$ and $h \leq_{\mathrm{UC}} k$ then $h \bullet f \leq_{\mathrm{UC}} k \bullet g$. With $s, t$ the dummy simulators of the two hypotheses, the simulator for $a$ is $a \circ (\mathrm{id} \otimes t) \circ (s \otimes \mathrm{id})$.
4. Regrading along a split monomorphism $c$ of $\mathcal{I}$ preserves $\leq_{\mathrm{UC}}$.

(Agda: ≤UC-refl, ≤UC-trans, ≤UC-resp-≈, ≈ᵁ⇒≤UC, dummy-complete, ≤UC⇒dummy, ≤UCᵈ⇔≤UC, UC-compose, ≤UC-sub)

*Proof idea.* (2) is the heart. Given $s_0$, the congruence of $\approx^{\mathrm{U}}$ under $\mathrm{sub}_a$ and functoriality of $\mathcal{M}$ give $\mathrm{sub}_a\, f \approx^{\mathrm{U}} \mathrm{sub}_a\, \mathrm{sub}_{s_0}\, g = \mathrm{sub}_{a s_0}\, g$; conversely take $a = \mathrm{id}$. (3) chains the congruences of Proposition 4 with the naturality of sub against ext.

These four statements are the UC metatheorems of the literature ([2]; [3]): every setup below inherits them, with the same proofs, because they are stated over $\approx^{\mathrm{U}}$ rather than over $\approx_{\mathcal{E}}$.

## 1.4 Morphisms of setups

**Definition 9** (setup morphism). A morphism $\mathbb{M} = (F, \Phi, \kappa, \nu) : \mathbb{S} \to \mathbb{S}'$ consists of

- a functor $F : \mathcal{C} \to \mathcal{C}'$;
- a lax monoidal functor $\Phi : \mathcal{I} \to \mathcal{I}'$, with structure maps $\Phi_\varepsilon : I' \to \Phi I$ and $\Phi_{X,Y} : \Phi X \otimes \Phi Y \to \Phi(X \otimes Y)$;
- components $\kappa_{X,A} : F(T_X A) \to T'_{\Phi X}(FA)$ satisfying

$$\kappa \circ F\eta_A = \mathrm{sub}'_{\Phi_\varepsilon} \circ \eta'_{FA}, \quad \kappa \circ F(\mathrm{ext}_X\, f) = \mathrm{sub}'_{\Phi_{X,Y}} \circ \mathrm{ext}'_{\Phi X}(\kappa \circ Ff) \circ \kappa, \quad \kappa \circ F(\mathrm{sub}_c) = \mathrm{sub}'_{\Phi c} \circ \kappa;$$

- a natural transformation $\nu : \mathcal{E} \Rightarrow \mathcal{E}' \circ F^{\mathrm{op}}$: a test on $A$ becomes a test on $FA$.

The *action on protocols* is $G(f) := \kappa \circ Ff : FA \to T'_{\Phi X}(FB)$. $\nu$ is *Epi* if every component $\nu_A$ is surjective, *Mono* if every component is injective (as maps of setoids). (Agda: UCSetupMorphism, GradedKleisliMorphism, IsGradedKleisliMorphism, PresheafMorphism, G, Epi, Mono)

The three $\kappa$-laws say that $(F, \kappa)$ is a morphism of graded Kleisli triples along the change of grading $\Phi$: $\kappa$ is a 2-cell between the two ways around the square $\mathcal{I} \times \mathcal{C} \to \mathcal{C}'$, compatible with unit and multiplication. The general constructions are all formal:

- *identities and composites* $\mathrm{id}_{\mathbb{S}}$ and $\mathbb{M}' \circ \mathbb{M}$ (with test map $\nu'_F \circ \nu$); the unit laws hold componentwise (Agda: idᵁ, composeᵁ, identityʳ-κ, identityˡ-κ)
- *change of kernel*: for $\nu : \mathcal{E} \Rightarrow \mathcal{E}'$ over the same $(\mathcal{C}, \mathcal{I}, \mathcal{M})$, $\mathsf{ck}_\nu := (\mathrm{Id}, \mathrm{Id}, \mathrm{id}, \nu)$ (Agda: changeKernel, reobserve)
- *image factorisation*: every $\mathbb{M}$ factors as an Epi change of kernel $\mathbb{S} \to \mathbb{S}[\mathrm{Im}\,\nu]$ followed by a morphism with Mono test map, where $\mathrm{Im}\,\nu(A)$ is $\mathcal{E}(A)$ with the equality induced by $\nu_A$ (Agda: Factorization, image, coarsen-to-image, image-morphism)
- *wide subcategories*: a wide subcategory $\mathcal{W} \subseteq \mathcal{C}$ closed under identities, composition, $\eta$, $\mu$, every $T_V$ and every $\mathrm{sub}_c$, with tests $\mathcal{E}'$ on $\mathcal{W}$ presented by $\nu : \mathcal{E}' \Rightarrow \mathcal{E} \circ \iota^{\mathrm{op}}$, gives a setup $\mathbb{S}|_{\mathcal{W}}$ and an inclusion $\mathsf{incl} : \mathbb{S}|_{\mathcal{W}} \to \mathbb{S}$; for $\mathcal{E}' = \iota^*\mathcal{E}$, emulation in $\mathbb{S}|_{\mathcal{W}}$ is the restriction of emulation in $\mathbb{S}$ (Agda: SetupSubcategory, Restriction, inclusion, restrict-≤UC, unrestrict-≤UC).

**Theorem 10** (transport along a morphism). Let $\mathbb{M} = (F, \Phi, \kappa, \nu) : \mathbb{S} \to \mathbb{S}'$.

1. If $\nu$ is Epi, $F$ preserves $\approx_{\mathcal{E}}$; if $\nu$ is Mono, $F$ reflects $\approx_{\mathcal{E}}$.
2. If $\nu$ is Epi and $\mathbb{S}'$ is grade-stable, then $f \le_{\mathrm{UC}} g \Rightarrow Gf \le_{\mathrm{UC}} Gg$; contrapositively, an impossibility proved in $\mathbb{S}'$ is one in $\mathbb{S}$.
3. If $\Phi$ is full, $\kappa$ is mono up to $\approx_{\mathcal{E}}$ ($\kappa \circ h \approx_{\mathcal{E}} \kappa \circ h' \Rightarrow h \approx_{\mathcal{E}} h'$), $\nu$ is Mono and $\mathbb{S}$ is grade-stable, then $Gf \le_{\mathrm{UC}} Gg \Rightarrow f \le_{\mathrm{UC}} g$.

(Agda: epi⇒preserves, mono⇒reflects, epi⇒preserves-≈ℰ, mono⇒reflects-≈ℰ, transfer, transfer-impossible, κ-Mono, reflects-≤UC)

*Proof idea.* (1) is one naturality square of $\nu$, cancelled on the surjective or the injective side. For (2): take the dummy simulator $s_0$ of $f \le_{\mathrm{UC}} g$; Epi carries $f \approx_{\mathcal{E}} \mathrm{sub}_{s_0} g$ along $F$; the $\kappa$-law for sub rewrites $G\big(\mathrm{sub}_{s_0} g\big)$ as $\mathrm{sub}'_{\Phi s_0} Gg$; grade stability of $\mathbb{S}'$ (Proposition 6) lifts the target $\approx_{\mathcal{E}}$ to $\approx^{\mathrm{U}}$; dummy completeness concludes.

The hypotheses of (2) and (3) fail in most of the interesting cases of Section 5: the main model is not grade-stable, and test maps out of syntactic models are rarely Epi. Two hypothesis-free transports are therefore used instead: the refinement below, and explicit characterisations of emulation such as the construction theorem of Section 3.2.

**Theorem 11** (refinement of the observation). Let $\mathcal{E}'$ be a second test presheaf on the same $\mathcal{C}$ with $\approx_{\mathcal{E}} \subseteq \approx_{\mathcal{E}'}$. Then $\approx^{\mathrm{U}} \subseteq \approx^{\mathrm{U}'}$ and $\le_{\mathrm{UC}} \subseteq \le'_{\mathrm{UC}}$, with the same simulators. No epi, mono or grade-stability hypothesis is spent. (Agda: Refine, ≈ᵁ-refine, ≤UC-refine)

## 1.5 The standard instantiation

**Definition 12** (standard setup). For a monoidal category $\mathbb{M}$ and a presheaf $\mathcal{E}$ on its underlying category, $\mathsf{Std}(\mathbb{M}, \mathcal{E}) := (|\mathbb{M}|, \mathbb{M}, X \mapsto X \otimes (-), \mathcal{E})$: computations and interfaces are the same category, and the graded monad is the curried tensor. (Agda: StdUC, StdSetup, 𝓜-standard, curriedTensor)

Concretely $T_X B = X \otimes B, T_W h = \mathrm{id}_W \otimes h, \eta = \lambda^{-1}, \mu_{X,Y} = \alpha^{-1} : X \otimes (Y \otimes B) \to (X \otimes Y) \otimes B, \mathrm{sub}_c = c \otimes \mathrm{id}$ and $h \bullet f = \alpha^{-1} \circ (\mathrm{id}_X \otimes h) \circ f$. Since $\alpha$ is invertible, the two relations of Section 1.3 read

$$f \approx^{\mathrm{U}} g \iff \forall W.\quad \mathrm{id}_W \otimes f \approx_{\mathcal{E}} \mathrm{id}_W \otimes g \quad \text{(agreement under every left ancilla)},$$
$$f \leq_{\mathrm{UC}} g \iff \forall a : X \to X'. \exists s : Y \to X'.\quad (a \otimes \mathrm{id}_B) \circ f \approx^{\mathrm{U}} (s \otimes \mathrm{id}_B) \circ g.$$

An adversary is an ordinary morphism plugged onto the adversary interface, a simulator one plugged onto the ideal protocol's. **Every setup in this note is standard.**

**Proposition 13** (standard morphisms). If $F : \mathbb{M} \to \mathbb{M}'$ is strong monoidal and $\nu : \mathcal{E} \Rightarrow \mathcal{E}' \circ F^{\mathrm{op}}$, then $(F, F, \delta, \nu)$ is a setup morphism $\mathsf{Std}(\mathbb{M}, \mathcal{E}) \to \mathsf{Std}(\mathbb{M}', \mathcal{E}')$, where $\delta_{X,A} : F(X \otimes A) \to FX \otimes FA$ is the inverse of the strong comparison. (Agda: `StdUCMorphism`)

The general Definition 9 relaxes "strong" only to "oplax relative to $\Phi$": $\kappa$ always points from $F(X \otimes A)$ to $\Phi X \otimes FA$. A lax crossing — one input producing several outputs, or several inputs folded into one — is never a setup morphism; this is the common root of several non-existence results in Section 5.3.

**Proposition 14** (monoidal quotients and the two-sided ancilla kernel).

1. Let $\sim$ be a *monoidal congruence* on $\mathbb{M}$: an equivalence on each hom-set that contains the hom-equality and is preserved by $\circ$ and $\otimes$. Then $\mathbb{M}/\sim$ (same objects, morphisms and structure, equality $\sim$) is monoidal, and the identity-on-the-nose functor $\mathbb{M} \to \mathbb{M}/\sim$ is strong monoidal. If moreover $\sim \subseteq \approx_{\mathcal{E}}$, then $\mathcal{E}$ descends to $\mathbb{M}/\sim$ and the comparison of tests is the identity, both Epi and Mono.
2. For any $\mathcal{E}$, the relation $f \sim^{\mathcal{E}} g :\iff \forall Y, Z.\, (\mathrm{id}_Y \otimes f) \otimes \mathrm{id}_Z \approx_{\mathcal{E}} (\mathrm{id}_Y \otimes g) \otimes \mathrm{id}_Z$ is a monoidal congruence (no braiding is needed: the other factor of a tensor is absorbed into an ancilla) lying between the hom-equality and $\approx^{\mathrm{U}}$; in particular it is contained in $\approx_{\mathcal{E}}$, so $\mathcal{E}$ descends to the quotient.

(Agda: `MonoidalCongruence`, `quotient`, `quotientF`, `descend`, `descend^v-mono`, `descend^v-epi`, `uKernel`, `~⇒≈ℰ`, `~⇒≈ᵘ`)

## 1.6 Robust compilation: a theorem about $\leq_{\mathrm{UC}}$, not a setup

Let $\mathsf{Pred}^{\mathrm{U}}(A, B)$ consist of families $H = (H_{X'})_{X'}$ of $\approx^{\mathrm{U}}$-invariant predicates on attacked protocols $A \to T_{X'} B$, and say $f$ *robustly satisfies* $H$ if $H(\mathrm{sub}_a \circ f)$ holds for every adversary $a$.

**Theorem 15** (UC is robust compilation).

1. $f \leq_{\mathrm{UC}} g$ iff every $H \in \mathsf{Pred}^{\mathrm{U}}(A, B)$ robustly satisfied by $g$ is robustly satisfied by $f$. The witness for "if" is $\mathrm{Hyp}(g) :=$ "is $\approx^{\mathrm{U}}$-equal to an attacked $g$".
2. Robust preservation of *environment-blind* predicates (on observations, ignoring the environment) is exactly *outcome inclusion* $f \overset{o}{\subseteq} g :\iff \forall W, e.\, \exists e'.\, \mathrm{run}_W(f)(e) \approx \mathrm{run}_W(g)(e')$; it is strictly weaker than $\leq_{\mathrm{UC}}$ (Theorem 50).
3. *Trace form.* $f \approx^{\mathrm{U}} g$ iff $f$ and $g$ robustly satisfy the same environment-indexed trace properties; hence $f \leq_{\mathrm{UC}} g$ iff for every $a$ there is $s$ such that every such property of $\mathrm{sub}_s \circ g$ is one of $\mathrm{sub}_a \circ f$.
4. *Two setups.* Along $\mathbb{M} : \mathbb{S} \to \mathbb{S}'$: if $\nu$ is Epi and $\mathbb{S}'$ grade-stable, $f \leq_{\mathrm{UC}} g$ implies that every robust hyperproperty of $Gg$ in $\mathbb{S}'$ is one of $Gf$; conversely if $\Phi$ is full, $\kappa$ mono up to $\approx_{\mathcal{E}}$, $\nu$ Mono and $\mathbb{S}$ grade-stable.

(Agda: `Pred^U`, `Robust^U`, `Hyp`, `≤UC⇔robust^U`, `_⊆°_`, `robust⇔outcomes`, `≈^U⇔robust^t`, `≤UC⇔robust^t`, `robust^U-transfer`, `robust^U-reflect`, `≤UC⇔robust^U-along`)

Item 1 is Patrignani et al.'s "UC is robust hyperproperty preservation" ([4], Thm 4; [5], Thm 3) in this setting; it needs neither grade stability nor any closure condition on environments. In item 3 the simulator is chosen before the property. Two refinements are designed but not formalised **DESIGN**: the *reactive-simulatability* order $\forall a \forall W, e \exists s.\, \mathrm{run}_W(\mathrm{sub}_a f)(e) \approx \mathrm{run}_W(\mathrm{sub}_s g)(e)$, which robust preservation of all trace properties should characterise (the conjecture of [4], §8.2.1) and which lies between $\leq_{\mathrm{UC}}$ and outcome inclusion; and the criteria hierarchy of robust compilation ([6]) as a family of test presheaves on one $(\mathcal{C}, \mathcal{I}, \mathcal{M})$, ordered by kernel refinement (Theorem 11).

## 2 The setups at a glance

All setups are standard (Definition 12), so each row gives the monoidal category $\mathbb{M}$ — objects with their tensor, morphisms with their hom-equality — and the test presheaf. DESIGN rows are planned, not built.

| Name | Objects; $\otimes$ | Morphisms; hom-equality | Tests $\mathcal{E}(A)$ | Model of | Status; source |
|---|---|---|---|---|---|
| PM<br>protocol model | interfaces $A = (A^+, A^-)$, pairs of sets; componentwise $+$ | probabilistic Mealy machines $A^+ + B^- \to A^- + B^+$ over the probabilistic delay monad, composed by feedback (G-construction); simulation equivalence $\approx_{\mathsf{M}}$ | tests $e : A \to \Omega$, identified when $\mathrm{Obs}(e \circ m) \sim^{\mathsf{a}} \mathrm{Obs}(e' \circ m)$ for every closure $m : 0 \to A$ | UC [3] | FORMAL `StdUC` $\mathbb{G}^{\mathsf{o}}$ $\mathcal{E}^{\mathsf{o}}$, $\mathbb{G}^{\mathsf{o}}$, $\mathcal{E}^{\mathsf{o}}$ |
| RS<br>random systems | as PM | PM's machines; behavioural equivalence $\approx_{\mathsf{R}}$: equal verdict laws against every adaptive distinguisher | PM's tests, descended | random systems [7], constructive cryptography [8] | FORMAL `StdUC` $\mathbb{G}^{\mathsf{R}}$ $\mathcal{E}^{\mathsf{R}}$, $\mathbb{G}^{\mathsf{R}}$, $\mathcal{E}^{\mathsf{R}}$ |
| $\mathcal{W}[\iota^*\mathcal{E}^o]$, $\mathcal{W}[\mathcal{E}^W]$<br>well-bracketed machines | oracle signatures (as SSP); tensor transported from PM | PM morphisms between flattened signatures that are $\approx_{\mathsf{M}}$-equal to a well-bracketed machine; $\approx_{\mathsf{M}}$ | $\iota^*\mathcal{E}^o$: all PM tests; $\mathcal{E}^W$: well-bracketed tests modulo well-bracketed closures | call/return discipline inside PM respectfulness [9] | FORMAL $\mathcal{W}^{\mathsf{M}}$, `StdUC.StdSetup` $\mathcal{W}^{\mathsf{M}}$ $\iota^*\mathcal{E}^{\mathsf{o}}$, `StdUC` $\mathcal{W}^{\mathsf{M}}$ $\mathcal{E}^{\mathsf{W}}$ |
| PM[wb]<br>well-bracketed quotient | as $\mathcal{W}$ | $\mathcal{W}$'s morphisms modulo the two-sided ancilla kernel $\sim_{\mathcal{W}}$ of well-bracketed tests | $\mathcal{E}^W$, descended | PM against call/return environments | FORMAL `StdUC` `PM[wb]`$^{\mathsf{M}}$ `E[wb]`, `PM[wb]`$^{\mathsf{M}}$, `E[wb]` |
| SSP<br>call/return packages | oracle signatures $(\mathsf{Ch}, Q, R)$; disjoint union of channels | packages: a state, an initial state, and a step answering each query by a finite tree of calls, rational coins and aborts; composition by inlining; $\approx_{\mathsf{p}}$: simulations up to syntactic tree equality | distinguishers $D : A \to \Omega$, identified when all games $G : 1 \to A$ give close verdict subdistributions | state-separating proofs [10], [11] | FORMAL `StdUC` `Prot`$^{\mathsf{M}}$ $\mathcal{E}^{\mathsf{p}}$, `Prot`$^{\mathsf{M}}$, $\mathcal{E}^{\mathsf{p}}$ |
| ITM<br>interactive Turing machines | finite trees of bitstring ports; concatenation of trees | terms of the free monoidal category on finite-table interactive Turing machines; free monoidal congruence | PM's tests of the interpreted interface | ITMs [3], IITM [12] | FORMAL `StdUC` $\mathbb{T}^{\mathsf{M}}$ $\mathcal{E}^{\mathsf{T}}$, $\mathbb{T}^{\mathsf{M}}$, $\mathcal{E}^{\mathsf{T}}$ |
| $\mathsf{ITM}_{\cong}$<br>ITMs, relational reading | as ITM | as ITM | Iso's tests of the relational interpretation | ITMs, possibilistically | FORMAL `StdUC` $\mathbb{T}^{\mathsf{M}}$ $\mathcal{E}^{\mathsf{Iso}}$, $\mathcal{E}^{\mathsf{Iso}}$ |
| Iso<br>relational machines | channels $(\mathsf{in}, \mathsf{out})$ of sets; $+$, grade on the right | relations $S \times \mathsf{In} \times (\mathsf{Out} + 1) \times S$ without initial state; composition by ping-pong; $\cong_{\mathsf{M}}$: step-preserving bijection of state sets | pairs $(Y, E : A \otimes Y \to \Omega)$ of an ancilla and an environment, up to $\cong_{\mathsf{M}}$ | nondeterministic machines up to isomorphism | FORMAL `StdUC machines` $\mathcal{E}$`-tests`, `machines`, $\mathcal{E}$`-tests` |
| Rel<br>relations | sets; $\times$ | relations; equality of relations | subsets, i.e. $\mathrm{Hom}(-, 1)$ | possibilistic one-shot security; cf. [13] | FORMAL `StdUC Rel` $\mathcal{E}$`-tests`, `Rel`, $\mathcal{E}$`-tests` |
| $\mathsf{PM}^{\omega}$<br>vanishing tier | $\mathbb{N}$-indexed families of PM interfaces; levelwise | levelwise PM morphisms certified at a rate polynomial in the security parameter; levelwise $\approx_{\mathsf{M}}$ | family tests, identified up to vanishing advantage | asymptotic UC | FORMAL `ucSetup^ω`, at PM: $\mathbb{G}^{\mathsf{o}}$ `qevaluation`$^{\mathsf{o}}$ `grading`$^{\mathsf{o}}$ |
| $\mathsf{PM}^{\mathsf{N}}$<br>negligible tier | as $\mathsf{PM}^{\omega}$ | as $\mathsf{PM}^{\omega}$ | family tests, identified up to local negligible advantage | computational UC [2] | FORMAL `ucSetup`$^{\mathsf{N}}$, at PM: $\mathbb{G}^{\mathsf{o}}$ `qevaluation`$^{\mathsf{o}}$ `grading`$^{\mathsf{o}}$ |
| QPM<br>quantitative PM | as PM | as PM | an approximation space: $e \approx_{\varepsilon} e'$ iff every closure is observed within $\varepsilon$ | PM with errors kept | FORMAL `QSetup`, at PM: `QSetup to QSetup`$^{\mathsf{o}}$ |
| QRS<br>quantitative RS | as RS | as RS | RS's tests at a measured error, read through the *closed* verdict slack | constructive cryptography with an exact $\varepsilon$ [8] | FORMAL `QSetup`$^{\mathsf{R}}$, `qevaluation`$^{\mathsf{R}}$ |

| Name | Objects; $\otimes$ | Morphisms; hom-equality | Tests $\mathcal{E}(A)$ | Model of | Status; source |
|---|---|---|---|---|---|
| $\mathsf{SSP}^{\mathsf{d}}$<br>distributional SSP | as SSP | SSP's packages under a distributional equality of call trees | as SSP | SSP with deferred sampling as an equation | **DESIGN** `examples-ssp-otp.md` |
| IPDL | polarised typed channel contexts pos $\rightleftharpoons$ neg; concatenation | reaction networks; composition new $B$ in $(f \parallel g)$; IPDL's exact equational theory | IPDL protocols into $\Omega$, identified modulo IPDL closures | IPDL [14] | **DESIGN** `uc-setup-ipdl.md` |
| CB<br>causal boxes | pairs of slot alphabets; $\times$ (concurrency) | classical causal boxes on positions $\mathbb{N}$, two phases per position; behavioural equality under adaptive environments | distinguishers (Portmann Def. 7.1) | causal boxes [15] | **DESIGN** `uc-setup-causal-boxes.md` |
| Sym, $\mathsf{Sym}_{\cong}$<br>Dolev–Yao | port trees of term sorts | free monoidal category on Dolev–Yao roles; free monoidal congruence | Dolev–Yao roles with an ancilla; Sym: possible verdicts under globally fresh naming (DY trace equivalence); $\mathsf{Sym}_{\cong}$: syntactic equality | symbolic security [16], [17] | **DESIGN** `uc-setup-symbolic.md` |
| $\mathsf{ND}_{\mathsf{A}}$, $\mathsf{ND}_{\mathsf{B}}$<br>nondeterministic choice | PM's (or SSP's) objects; grades $X \otimes C_{S_1} \otimes \cdots \otimes C_{S_n}$ with choice factors $C_S = (1, S)$ | PM's morphisms; grade morphisms: all of them between such grades ($\mathsf{ND}_{\mathsf{B}}$), or $u \otimes \rho$ with $\rho$ a relabelling of choice factors ($\mathsf{ND}_{\mathsf{A}}$) | the ambient tests | B: adversary-resolved scheduling [3]; A: environment-resolved specifications | **DESIGN** `uc-setup-nondeterministic.md` (superseded design, Section 4.5) |

# 3 The setups

## 3.1 PM: the protocol model

**The base.** $D_p$ is the rational probabilistic delay monad: $D_pA = \nu X.S_{\mathbb{Q}}(A + X)$, coinductive trees of biased rational coins whose leaves are values or further delay; mass that never lands is divergence, and $\approx_p$ is the delay-insensitive equality of the resulting laws, whose masses are lower reals (Agda: `Dₚ`, `_≈ₚ_`). Its Kleisli category on sets is symmetric monoidal under $\times$ (commutativity of $D_p$), distributive, and has an Elgot iteration (Agda: $\mathcal{U}_\mathtt{p}$, `Elgotₚ`). A *Mealy machine* $A \to B$ over it is a state set $S$, an initial law $s_0 \in D_pS$ and a step $S \times A \to D_p(S \times B)$; machines form a symmetric monoidal category under $+$, traced by the iteration-based $+$-trace (Agda: `Machine`, `State`, $\mathcal{M}_\mathtt{p}$, `Tracedₚ`) (cf. [18], [19]). Their hom-equality is *simulation equivalence* $\approx_{\mathsf{M}}$: the equivalence closure of the relation "there is a function $\theta : S \to S'$ with $\theta s_0 = s_0'$ and $\mathrm{step}' \circ (\theta \times \mathrm{id}) = (\theta \times \mathrm{id}) \circ \mathrm{step}$" (Agda: `_≲_`, `_≈ᴹ_`).

**Definition 16** (PM). $\mathbb{G}^o$ is the Int (G) construction ([20]) on traced Mealy machines. Its objects are interfaces $A = (A^+, A^-)$ ($A^+$ travels up to the caller, $A^-$ down); a morphism $A \to B$ is a machine $A^+ + B^- \to A^- + B^+$; composition is feedback over the middle interface, the identity is the copycat, and $A \otimes C = (A^+ + C^+, A^- + C^-)$. The verdict interface is $\Omega = (\mathsf{Bool}, 1)$: the environment's single activation is the tick on 1. A *closure* of $A$ is a morphism $m : 0 \to A$; a closed system $u : 0 \to \Omega$ is observed as $\mathrm{Obs}(u) \in D_p\ \mathsf{Bool}$, its run on the tick. Two laws are close, $d \sim^{\mathsf{a}} d'$, when for every rational $\varepsilon > 0$ their $\mathsf{true}$-masses and their $\mathsf{false}$-masses agree within $\varepsilon$ (comparing both masses separates answering $\mathsf{false}$ from diverging). The tests are

$$\mathcal{E}^o(A) = \mathbb{G}^o(A, \Omega)/\!\approx, \quad e \approx e' :\Longleftrightarrow \forall m : 0 \to A.\, \mathrm{Obs}(e \circ m) \sim^{\mathsf{a}} \mathrm{Obs}(e' \circ m),$$

the image of the evaluation functional, and $\mathsf{PM} := \mathsf{Std}(\mathbb{G}^o, \mathcal{E}^o)$. (Agda: $\mathcal{G}_\mathtt{p}{}^{\mathtt{M}}$, $\mathbb{G}^\mathtt{o}$, $\Omega^\mathtt{o}$, `Obs`, `Approximationᴹ`, `Evaluation`, $\mathcal{E}^\mathtt{o}$, `evaluationᵒ`, $\mathcal{E}^\mathtt{o}$, `StdUC` $\mathbb{G}^\mathtt{o}$ $\mathcal{E}^\mathtt{o}$)

In PM, $f \approx_{\mathcal{E}} g$ says that every closed context $t \circ f \circ m$ is observed like $t \circ g \circ m$; $f \approx^{\mathrm{U}} g$ says the same with an arbitrary ancilla beside the hole; and $f \lesssim_{\mathrm{UC}} g$ is UC emulation in the usual "for every adversary there is a simulator such that no environment tells them apart" form, the environment being a pair (closure, test). Compared with [2], machines are arbitrary probabilistic Mealy machines rather than polynomial ITMs, and the execution model is the one of the G-construction: each activation is answered by exactly one message, and divergence is missing mass. The hom-equality is observed exactly: $u \approx_{\mathsf{M}} v \Rightarrow \mathrm{Obs}(u) \approx_p \mathrm{Obs}(v)$ (Agda: `obs-resp`).

**Theorem 17** (full abstraction of PM). For $f, g : A \to X \otimes B$ in PM, $f \approx^{\mathrm{U}} g$ iff $f \approx_{\mathsf{R}} g$, i.e. iff $f$ and $g$ are the same random system (Section 3.2). One ancilla suffices: if $T_{A^*} f \approx_{\mathcal{E}} T_{A^*} g$ for the dual interface $A^* = (A^-, A^+)$, then $f \approx_{\mathsf{R}} g$. (Agda: `≈ᵁ⇒≈ᴿ`, `T₁-≈ℰ⇒∼ᴿ`, `≈ᴿ⇒≈ᵁ`)

*Proof idea.* ($\Rightarrow$) The closure $\mathrm{cup} : 0 \to A^* \otimes A$ hands the letters of a distinguisher strategy $d$ to the hole and back, and the test that plays $d$ observes exactly the run of $d$ against $f$: $\mathrm{Obs}((\mathrm{play}_d \circ (\mathrm{id} \otimes f)) \circ \mathrm{cup}) \approx_p \mathrm{run}(f, d)$ (Agda: `play-cup`, `cupᵒ`, `playᵒ`). ($\Leftarrow$) The test map of $\mathsf{rs} : \mathsf{PM} \to \mathsf{RS}$ is Mono (Section 5.1).

So the library's security notion has a semantic meaning: two protocols are UC-equivalent in PM exactly when they are the same random system in the sense of [7].

**Proposition 18** (PM is not grade-stable). Let $A = (\emptyset, \mathsf{Bool})$, $B = (\emptyset, 1)$ and let $r_b : A \to B$ be the stateless machine that answers the activation from $B$ by sending $b$ down to $A$. Then $r_{\mathsf{true}} \approx_{\mathcal{E}} r_{\mathsf{false}}$ — a closure of $A$ can receive $b$ but has nothing to answer with, so every closed composite diverges — while $r_{\mathsf{true}} \not\approx_{\mathsf{R}} r_{\mathsf{false}}$, hence (Theorem 17) $T_{A^*} r_{\mathsf{true}} \not\approx_{\mathcal{E}} T_{A^*} r_{\mathsf{false}}$. So PM is not grade-stable. (Agda: `revealᴹ`, `reveal-≈ℰ`, `reveal-≉ᴿ`, `¬gradeStableᵒ`)

Consequently Proposition 6(i) is unavailable in PM: a fact proved at $\approx_{\mathcal{E}}$ reaches $\approx^{\mathrm{U}}$ through behavioural equality (Theorem 17) instead, and the target hypothesis of Theorem 10(2) never holds with PM as target. Emulation in PM is invariant under behavioural equality of either side (Agda: `≲UC-resp-≈ᴿ`).

## 3.2 RS: random systems

**Definition 19** (distinguishers, behavioural equivalence). A *distinguisher* for a machine with inputs $X$ and outputs $Y$ is a finite decision tree $d \in \mathsf{Strat}(X, Y)$ whose leaves output a bit and whose nodes either ask $x \in X$ and branch on the answer, or toss a rational coin; $\mathrm{run}(f, d) \in D_p\ \mathsf{Bool}$ runs $f$ from its initial state against $d$. Then $f \approx_{\mathsf{R}} g$ iff $\mathrm{run}(f, d) \sim^{\mathsf{a}} \mathrm{run}(g, d)$ for every $d$ (adaptive equivalence, [7] Defs. 9–10). The non-adaptive equivalence $f \approx_{\mathsf{B}} g$ asks, for every input word and every $[0,1]$-valued function on output words, equal expectations within every $\varepsilon > 0$ ([7] Def. 3). (Agda: `Strat`, `runˢ`, `_≈ᴿ_`, `_≈ᴮ_`)

**Theorem 20** (behavioural equivalence is a monoidal congruence). $\approx_{\mathsf{M}} \subseteq \approx_{\mathsf{R}}$, and $\approx_{\mathsf{R}}$ is preserved by composition and tensor of $\mathbb{G}^o$ ([7] Lemma 4). (Agda: `≈ᴹ⇒≈ᴿ`, `compose⁶-resp-≈ᴿ`, `⊗₁⁶-resp-≈ᴿ`, `congᴿ`)

*Proof idea.* A context that a distinguisher can simulate is *absorbed* into it: the context's own coins are unfolded into the strategy, and since the context may loop while strategies are finite, the absorbed distinguisher is the family of its budget-truncations (Agda: `Absorbs`, `absorbs⇒resp`). Composition and tensor with a fixed machine on either side are absorbable.

**Definition 21** (RS). $\mathbb{G}^R := \mathbb{G}^o / \approx_{\mathsf{R}}$ (Proposition 14). Tests cannot separate behaviourally equal machines, so $\mathcal{E}^o$ descends to $\mathcal{E}^R$, and $\mathsf{RS} := \mathsf{Std}(\mathbb{G}^R, \mathcal{E}^R)$. (Agda: `𝔾ᴿ`, `~ᴿ⇒≋`, `ℰᴿ`, `StdUC 𝔾ᴿ ℰᴿ`)

**Proposition 22** (non-adaptive suffices on finite alphabets). If the query alphabet $X$ is finite with decidable equality, $f \approx_{\mathsf{B}} g \Rightarrow f \approx_{\mathsf{R}} g$. (Agda: `≈ᴮ⇒≈ᴿ-fin`)

The general statement is classical ([9], Thm. 2) and open here; the adaptive $\approx_{\mathsf{R}}$ was chosen as the hom-equality because the congruence proof needs it.

**The random-systems toolkit.** A verdict mass is a lower real, so the distinguishing advantage is a relation in a rational slack rather than a number.

**Definition 23** (advantage). $\Delta_d(f,g) \le \varepsilon$ iff for every $\delta > 0$ the laws $\mathrm{run}(f,d)$ and $\mathrm{run}(g,d)$ have both masses within $\varepsilon + \delta$ (a *closed* slack); $f \approx^{\varepsilon}_{\mathsf{R}} g$ iff $\Delta_d(f,g) \le \varepsilon$ for every distinguisher $d$ ([7] Def. 10, [8] §4.5). (Agda: `Δ[_]⟨_,_⟩≤_`, `_≈ᴿ[_]_`)

**Proposition 24** ($\Delta$ is a pseudometric on RS). $f \approx^{\varepsilon}_{\mathsf{R}} g$ and $g \approx^{\varepsilon'}_{\mathsf{R}} h$ give $f \approx^{\varepsilon+\varepsilon'}_{\mathsf{R}} h$ ([7] Lemma 5(i)); $f \approx_{\mathsf{R}} g$ iff $f \approx^{0}_{\mathsf{R}} g$ iff $f \approx^{\varepsilon}_{\mathsf{R}} g$ for every $\varepsilon > 0$; and bounds are invariant under $\approx_{\mathsf{R}}$ on either side, so $\Delta$ descends to the homs of $\mathbb{G}^R$. (Agda: `≈ᴿ[]-trans`, `≈ᴿ⇔≈ᴿ[0]`, `≈ᴿ⇔≈ᴿ[+]`, `≈ᴿ[]-resp`)

**Theorem 25** (composition theorem at a slack). Absorbable contexts do not increase $\Delta$: $f \approx^{\varepsilon}_{\mathsf{R}} g \Rightarrow Cf \approx^{\varepsilon}_{\mathsf{R}} Cg$. Hence $f \approx^{\varepsilon}_{\mathsf{R}} h$ and $g \approx^{\delta}_{\mathsf{R}} i$ give $f \circ g \approx^{\varepsilon+\delta}_{\mathsf{R}} h \circ i$ and $f \otimes g \approx^{\varepsilon+\delta}_{\mathsf{R}} h \otimes i$ ([7] Lemma 5(ii)–(iii), [8] Def. 2), and a closing context observes at most $\Delta$: if $f \approx^{\varepsilon}_{\mathsf{R}} g$ then $\mathrm{Obs}((t \circ (\mathrm{id} \otimes f)) \circ m)$ and $\mathrm{Obs}((t \circ (\mathrm{id} \otimes g)) \circ m)$ are within $\varepsilon$ (closed slack). (Agda: `absorbs-Δ`, `absorbs-≈ᴿ[]`, `compose⁶-≈ᴿ[]`, `⊗₁⁶-≈ᴿ[]`, `∘-resp-~ᴿ[]`, `⊗-resp-~ᴿ[]`, `close-~ᴿ[]`)

**Lemma 26** (hybrid argument). In any approximation with rational errors, $H_i \approx_{\varepsilon_i} H_{i+1}$ for $i < n$ gives $H_0 \approx_{\sum_{i<n} \varepsilon_i} H_n$. In RS, if hybrid $i$ is $C_i(f)$ and hybrid $i+1$ is $C_i(g)$ up to $\approx_{\mathsf{R}}$, with every $C_i$ absorbable, then $f \approx^{\varepsilon}_{\mathsf{R}} g$ gives $H_0 \approx^{n\varepsilon}_{\mathsf{R}} H_n$. (Agda: `hybrid`, `hybrid-absorbs`)

**Definition 27** (construction). $f$ *constructs* $g$ within $\varepsilon$, $\mathsf{Constructs}_{\varepsilon}(f,g)$, iff there is a simulator $s$ with $f \approx^{\varepsilon}_{\mathsf{R}} \mathrm{sub}_s \circ g$ — the security condition of constructive cryptography ([8] §5.1, Def. 3), with every strategy as distinguisher. (Agda: `Constructs`)

**Theorem 28** (construction is emulation).

1. For $f, g$ in PM: $f \le_{\mathrm{UC}} g$ in PM $\iff \mathsf{Constructs}_0(f,g) \iff \mathsf{rs}(f) \le_{\mathrm{UC}} \mathsf{rs}(g)$ in RS.
2. $\Delta$ is the quantitative U-agreement of QPM (Section 3.8) up to slack: $f \approx^{\varepsilon}_{\mathsf{R}} g$ iff $f \approx^{\mathrm{U}}_{\varepsilon+\delta} g$ for every $\delta > 0$. Accordingly $\mathsf{Constructs}_{\varepsilon}(f,g)$ iff one simulator witnesses $\approx^{\mathrm{U}}_{\varepsilon+\delta}$ for every $\delta > 0$, and a quantitative witness at $\varepsilon$ is a construction within $\varepsilon$.
3. ([8] Thm. 1) $\mathsf{Constructs}_0(f,f)$; $\mathsf{Constructs}_{\varepsilon}(f,g)$ and $\mathsf{Constructs}_{\varepsilon'}(g,h)$ give $\mathsf{Constructs}_{\varepsilon+\varepsilon'}(f,h)$ with the simulators composed; $\mathsf{Constructs}_{\varepsilon}(f,g)$ and $\mathsf{Constructs}_{\varepsilon'}(u,v)$ give $\mathsf{Constructs}_{\varepsilon'+\varepsilon}(u \bullet f, v \bullet g)$.
4. In RS, $\approx^{\mathrm{U}}$ is $\approx_{\mathsf{R}}$.

(Agda: `≤UC⇔Constructs₀`, `≤UCᴿ⇔Constructs₀`, `~ᴿ[]⇔≈ᵁ[]`, `Witness⁺`, `Constructs⇔Witness⁺`, `Witness⇒Constructs`, `constructs-refl`, `constructs-trans`, `constructs-•`, `≈ᵁᴿ⇒~ᴿ`)

Not covered **DESIGN**: constructive cryptography's availability condition (a separate statement, since $\le_{\mathrm{UC}}$ never fixes the simulator of the silent adversary); parallel composition of constructions, which needs the interchange $(X \otimes B) \otimes (X' \otimes B') \cong (X \otimes X') \otimes (B \otimes B')$, i.e. a braiding (the G-construction on a symmetric traced category has one, but the formalised $\mathbb{G}^o$ is only monoidal); and amplification ([21]), which needs conditional probabilities and minima of masses that lower reals do not form. The exact-$\varepsilon$ version of Theorem 28 is the quantitative setup QRS (Section 3.8.2).

## 3.3 $\mathcal{W}$ and PM[wb]: well-bracketed machines

These two setups sit inside PM and restrict it to the call/return discipline of SSP (Section 3.4). For an oracle signature $A$ (Definition 34), its *flattening* flat $A$ is the PM interface whose upward messages are channel-tagged replies $\Sigma_c R_A(c)$ and whose downward messages are channel-tagged queries $\Sigma_c Q_A(c)$.

**Definition 29** (well-bracketed machine). A machine $f : \text{flat } A \to \text{flat } B$ is at any time in one of three *phases*: idle; serving $b$ (a query on $b \in \mathsf{Ch}_B$ is open); calling$(b, a)$ (it called $a \in \mathsf{Ch}_A$ on behalf of $b$). The environment's legal moves are a query on $b$ when idle (to serving $b$) and a reply on $a$ when calling$(b, a)$ (to serving $b$); the machine's legal moves from serving $b$ are a call on some $a$ (to calling$(b, a)$) and an answer on $b$ (to idle). $f$ is *well-bracketed* if some invariant on (state, phase) holds initially at idle and, from any state satisfying it, every legal environment move leads — on the support of the step, at every finite depth — to a legal move of $f$ into a state satisfying it again. (Agda: `Phase`, `Opp`, `Ply`, `W`)

This is CryptHOL's respectfulness ([9], Def. 3) moved from oracles to machines.

**Theorem 30** (closure of well-bracketedness). Compiled SSP packages are well-bracketed; so are the identity (copycat) and every relabelling wire that returns on the queried channel. Well-bracketed machines are closed under the composition of PM (collapsed to a single traced machine, which is $\approx_{\mathsf{M}}$-equal to the composite) and under the channel-dispatching tensor, which is $\approx_{\mathsf{M}}$-equal to PM's tensor conjugated by the relabellings $\varphi : \text{flat } A \otimes \text{flat } C \cong \text{flat}(A \otimes C)$. (Agda: `morphism-W`, `id-W`, `wire-W`, `⊙-W`, `⊙≈∘`, `⊗ᵂ-W`, `conj≈⊗ᵂ`)

**Definition 31** ($\mathcal{W}$, $\mathcal{W}[\iota^*\mathcal{E}^o]$, $\mathcal{W}[\mathcal{E}^W]$). $\mathcal{W}$ has oracle signatures as objects; $\mathcal{W}(A, B)$ consists of the PM morphisms $\text{flat } A \to \text{flat } B$ that are $\approx_{\mathsf{M}}$-equal to a well-bracketed machine; equality is $\approx_{\mathsf{M}}$; the monoidal structure is PM's transported along the relabelling isomorphisms $\varphi$ and $\upsilon : 0 \cong \text{flat } 1$. The inclusion $\iota : \mathcal{W} \to \mathbb{G}^o$ is strong monoidal, neither full nor wide. Two observations: $\mathcal{W}[\iota^*\mathcal{E}^o] := \mathsf{Std}(\mathcal{W}, \mathcal{E}^o \circ \iota^{\mathrm{op}})$ (all PM tests and closures), and $\mathcal{W}[\mathcal{E}^W] := \mathsf{Std}(\mathcal{W}, \mathcal{E}^W)$ with

$$\mathcal{E}^W(A) = \mathcal{W}(A, \Omega)/\!\approx\!\!, \quad t \approx t' :\Longleftrightarrow \forall m \in \mathcal{W}(1, A).\ \mathrm{Obs}(t \circ m) \sim^{\mathsf{a}} \mathrm{Obs}(t' \circ m),$$

well-bracketed tests modulo well-bracketed closures, observed exactly as PM observes them. (Agda: `Wᵒ`, `Uᵂ`, `φ`, `υ`, $\mathcal{W}^{\mathsf{M}}$, `ι`ᴹ, `ι*Eᵒ`, `Obsᵂ`, $\mathcal{E}^{\mathsf{W}}$, `StdUC` $\mathcal{W}^{\mathsf{M}}$ $\mathcal{E}^{\mathsf{W}}$)

Well-bracketedness reads the support of a step, which the equality $\approx_p$ of laws does not preserve; this is why $\mathcal{W}$ takes the saturation "equal to a well-bracketed machine" as its morphisms.

**Definition 32** (PM[wb]). Let $\sim_{\mathcal{W}}$ be the two-sided ancilla kernel of $\mathcal{E}^W$ (Proposition 14(2)). Then $\mathsf{PM}[\mathsf{wb}] := \mathsf{Std}(\mathcal{W}/\sim_{\mathcal{W}}, \mathcal{E}^W$ descended). (Agda: `PM[wb]`ᴹ, `~⇒≋`ᵂ, $\mathcal{E}$`[wb]`, `StdUC PM[wb]`ᴹ $\mathcal{E}$`[wb]`)

The quotient changes neither the contexts nor the kernel, so emulation in PM[wb] is emulation in $\mathcal{W}[\mathcal{E}^W]$. It is needed because the functor of a setup morphism must respect the *target's* hom-equality: with $\approx_{\mathsf{M}}$ the compiled identity wire of SSP is not the identity of PM (Section 5.3), while it is expected to be $\sim_{\mathcal{W}}$-equal to it (Section 5.2).

**Proposition 33** ($\iota$ carries the kernel both ways). $\iota : \mathcal{W}[\iota^*\mathcal{E}^o] \to \mathsf{PM}$ is a standard morphism whose test map is the identity, Epi and Mono. Hence $f \approx_{\mathcal{E}} g$ in $\mathcal{W}[\iota^*\mathcal{E}^o]$ iff $\iota f \approx_{\mathcal{E}} \iota g$ in PM. (Agda: `ι⇒prot`, `νι-epi`, `νι-mono`)

**DESIGN** The two observations on $\mathcal{W}$ are incomparable: neither is a sub-presheaf of the other, and no natural map exists in either direction. Two well-bracketed tests that differ only in how they react to a reply on the wrong channel agree on every well-bracketed closure but not on an undisciplined one; and "monitoring" an arbitrary test is not natural. So PM[wb] cannot be reached from $\mathcal{W}[\iota^*\mathcal{E}^o]$ by a change of kernel or an image factorisation (Section 1.4); the quotient morphism $\mathsf{q} : \mathcal{W}[\mathcal{E}^W] \to \mathsf{PM}[\mathsf{wb}]$ is designed (Section 5.2).

## 3.4 SSP: call/return packages

**Definition 34** (oracle signatures). An *oracle signature* $A = (\mathsf{Ch}_A, Q_A, R_A)$ is a set of channels with decidable equality, each with a query type $Q_A(c)$ and a reply type $R_A(c)$. $A \otimes C$ has channels $\mathsf{Ch}_A + \mathsf{Ch}_C$; the unit has none. A morphism of signatures $A \Rightarrow B$ maps channels and queries forwards and replies backwards (a container morphism). (Agda: `Iface`, `_⊗ᴵ_`, `_⇒ᴵ_`, `flat`)

**Definition 35** (packages). $\mathsf{Calls}_A(X)$ is the set of finite trees with leaves $\mathrm{ret}\, x$ and dead (abort) and nodes $\mathrm{call}\, c\, q$ (branching on $R_A(c)$) and $\mathrm{coin}\, \mu$ (a rational law on Bool). A *package* $P : A \to B$ — it imports $A$ and exports $B$ — is a state set St, an initial state, and

$$\mathrm{step} : \mathrm{St} \to (c : \mathsf{Ch}_B) \to Q_B(c) \to \mathsf{Calls}_A(\mathrm{St} \times R_B(c)).$$

$P_2 \circ P_1$ inlines $P_1$'s steps at the calls of $P_2$ ([10], Def. 4); the identity is the wire forwarding each query; $P \otimes P'$ serves a query with the factor owning its channel and replies on that channel, so control stays one thread; shared imports are $(P \otimes P') \circ \ulcorner\nabla\urcorner$. Equality $\approx_{\mathsf{p}}$ is the equivalence closure of *simulations*: functions $\theta : \mathrm{St} \to \mathrm{St}'$ preserving the initial state and carrying each step tree to the corresponding one, trees compared syntactically ($\approx_c$). This is a symmetric monoidal category. (Agda: `Calls`, `Protocol`, `_∘ᴾ_`, `wireᴾ`, `_⊗ᴾ_`, `_≈ᶜ_`, `_≲ᴾ_`, `_≈ᴾ_`, `Prot`, `Protᴹ`, `Protˢ`)

**Definition 36** (SSP). $\Omega$ has one channel with query 1 and reply Bool. A *game* is a package $G : 1 \to A$, a *distinguisher* one $D : A \to \Omega$. A closed $W : 1 \to \Omega$ is read as the finite subdistribution $\mathrm{run}(W) \in \mathsf{Dist}_{\perp}(\mathsf{Bool})$ obtained by multiplying out the coins (aborts are missing mass); laws are compared on both verdict masses. Then

$$\mathcal{E}^{p}(A) = \mathsf{SSP}(A, \Omega)/\approx, \quad D \approx D' :\Longleftrightarrow \forall G : 1 \to A.\, \mathrm{run}(D \circ G) \sim^{\mathsf{a}} \mathrm{run}(D' \circ G),$$

and $\mathsf{SSP} := \mathsf{Std}(\mathsf{Prot}, \mathcal{E}^{p})$. So $\approx_{\mathcal{E}}$ is perfect indistinguishability in every game context ([10] Def. 8 at $\varepsilon = 0$). (Agda: `evalC`, `runᵖ`, `Approximation⊥`, `evaluationᵖ`, `ℰᵖ`, `StdUC Protᴹ ℰᵖ`)

In the grade reading a protocol $A \to X \otimes B$ also exports adversary oracles $X$, and a simulator is a package plugged on them. The flat interface of PM is the one-channel case; channels are what lets the tensor answer each query on the channel it was asked on.

**Theorem 37** (state-separating reasoning). Write $G_0 \approx_{\varepsilon} G_1$ when every distinguisher reads the two games within $\varepsilon$. Then $G_0 \approx_{\varepsilon} G_1 \approx_{\delta} G_2 \Rightarrow G_0 \approx_{\varepsilon+\delta} G_2$ ([10] Lemma 5); $\mathrm{run}((D \circ M) \circ G) = \mathrm{run}(D \circ (M \circ G))$ (the reduction lemma, Lemma 6); and $G_0 \approx_{\varepsilon} G_1 \Rightarrow M \circ G_0 \approx_{\varepsilon} M \circ G_1$. (Agda: `_≈ᵍ[_]_`, `≈ᵍ-trans`, `reduction`, `≈ᵍ-reduce`)

**Proposition 38** (closed bridge). For *closed* protocols $f, g : 1 \to X \otimes B$, $f \approx_{\mathcal{E}} g \Rightarrow f \approx^{\mathrm{U}} g$, without grade stability; in particular perfectly indistinguishable games are U-equivalent. (Agda: `closed-bridge`, `≈ᵍ⇒≈ᵁ`)

*Proof idea.* An ancilla closure $m : 1 \to Y \otimes 1$ slides past a closed protocol by naturality, interchange and $\lambda_1 = \rho_1$, leaving one test of the protocol alone. The argument is valid in any standard setup whose closures start at the unit.

**Theorem 39** (the one-time pad). In SSP let aut be the authenticated channel (Eve reads the message and cannot change it), ideal the secure channel (Eve learns only that a message was sent), key a uniform key drawn when first fetched, and otp the encryption and decryption converters sharing the key, composed with aut and key ([8] §3.2–3.3; [10] Def. 9). Then $\mathrm{otp} \leq_{\mathrm{UC}} \mathrm{ideal}$, with the simulator "on Eve's first poll after a send, draw a uniform ciphertext and keep it". Moreover $\mathrm{ideal} \leq_{\mathrm{UC}} \mathrm{aut}$ and $\mathrm{otp} \leq_{\mathrm{UC}} \mathrm{aut}$ (leakage may be added), while $\mathrm{aut} \underset{\mathrm{UC}}{\not\leq} \mathrm{ideal}$. (Agda: `aut`, `ideal`, `key`, `otp`, `sim`, `otp≤ideal`, `ideal≤aut`, `otp≤aut`, `relay≤aut`, `leaky-not-secure`)

*Proof idea.* otp inlines to a single package real, and the simulated ideal to a package simulated, each by one simulation; real and simulated are perfectly indistinguishable by a deferred-sampling induction over the distinguisher, with the bijection $k \mapsto m \oplus k$ at Eve's poll; Proposition 38 turns that into $\approx^{\mathrm{U}}$, and dummy completeness concludes (Agda: `otp≤real`, `ideal≤simulated`, `perfect`). For $\mathrm{aut} \underset{\mathrm{UC}}{\not\leq} \mathrm{ideal}$: relabelling the message by negation is invisible to the secure channel and turns one probe into the other, whatever the simulator.

**Proposition 40** (no tree simulation). For every simulator $s$, $\mathrm{otp} \not\approx_{\mathsf{p}} \mathrm{sub}_s \circ \mathrm{ideal}$: simulations compare coins syntactically, and the key is drawn at Alice's send, whereas the simulated ideal can draw only at Eve's poll. (Agda: `no-tree-simulation`)

So the call/return discipline buys clean composition, not clean probabilistic reasoning: a deferred-sampling step is never a hom-equality of SSP and has to be proved at the level of runs. A distributional hom-equality on call trees is the design lever $\mathsf{SSP}^{\mathsf{d}}$ (Section 4) **DESIGN**.

## 3.5 ITM and $\mathsf{ITM}_{\cong}$: interactive Turing machines

**Definition 41** (ITM). A *port interface* is a finite binary tree whose leaves are ports of polarity $+$ (answers going out) or $-$ (queries coming in), each carrying bitstrings. A generator $A \to B$ is a finite-table Turing machine reading the ports $A^{+} + B^{-}$ and writing $A^{-} + B^{+}$: finitely many control states, a work tape that persists across activations, an input tape holding the activation's message and an output buffer; its instructions write and move, toss a fair coin, or *send* on a port, which ends the activation. $\mathbb{T}$ is the free monoidal category on these generators: morphisms are terms built from generators, identities, composition, tensor and the structural isomorphisms, modulo the free monoidal congruence $\approx_{\mathrm{Term}}$. Its interpretation $F^{p} : \mathbb{T} \to \mathbb{G}^{o}$ is the strict monoidal free functor whose leaves run their table: an activation iterates the table lookup (Elgot iteration) until it sends, coins are fair coins, and an activation that never sends diverges. $\mathsf{ITM} := \mathsf{Std}(\mathbb{T}, \mathcal{E}^{o} \circ (F^{p})^{\mathrm{op}})$: a test on ports is a PM test on their interpretation. (Agda: `Ports`, `Instr`, `TM`, `Gen`, `𝕋ᴹ`, `_≈Term_`, `runᵀᴹ`, `runᵖ`, `Fᵖ`, `freeStrongMonoidal`, `ℰᵀ`, `StdUC 𝕋ᴹ ℰᵀ`)

A system is a term, never a product table, so $F^{p}(g \circ f) = F^{p}g \circ F^{p}f$ holds by definition. This is the ITM model of [3] in the ports reading (one output per activation is the discipline of the G-construction), and of IITM ([12]); GNUC ([22]) and iUC ([23]) are restrictions of it by change of kernel or wide subcategory **DESIGN**.

**Definition 42** ($\mathsf{ITM}_{\cong}$). $F^{r} : \mathbb{T} \to \mathsf{Iso}$ is the strict monoidal free functor into the relational machines of Section 3.6 (whose tensor is reversed, so trees are read mirrored) that reads a leaf as its *halting relation*: from memory $m$ and

input $x$ the activation can end with output $y$ in memory $m'$; coins become nondeterministic choice, divergence has no transition, and the initial memory is forgotten. $\mathsf{ITM}_{\cong} := \mathsf{Std}(\mathbb{T}, \mathcal{E}^{\cong} \circ (F^r)^{\mathrm{op}})$. (Agda: `Halts`, `leafᴿ`, `Fᴿ`, `ℰᴵˢᵒ`, `StdUC 𝕋ᴹ ℰᴵˢᵒ`)

$\mathbb{T}$ carries two observations because Iso compares tests up to bijections of state sets, which no map out of the classes of $\mathcal{E}^o$ respects **DESIGN**.

**Theorem 43** (the kernel of ITM is that of PM). For parallel $f, g$ in $\mathbb{T}$: $f \approx_{\mathcal{E}} g$ in ITM iff $F^p f \approx_{\mathcal{E}} F^p g$ in PM. (Agda: `≈ℰ⇔`)

**DESIGN** Polynomial running time is not a property of morphisms: a priori PPT machines have no identities, and reactive PPT is not closed under composition ([24], §1.5). "Feasible" must therefore be a side condition over a security-parameter tier, not a sub-setup.

## 3.6 Iso: relational machines up to isomorphism

**Definition 44** (Iso). Objects are *channels* $(\mathsf{in}, \mathsf{out})$ of sets, with componentwise disjoint union. A machine $A \to B$ is a state set $S$ and a step relation $S \times \mathsf{In} \times (\mathsf{Out} + 1) \times S$ on the transposed channel: nondeterministic, possibly silent, and without initial state. Composition lets internal messages bounce between the two machines (an inductive ping-pong relation) until an external output or silence. The hom-equality $\cong_{\mathsf{M}}$ is a bijection of state sets that preserves and reflects steps. The tensor is reversed, so the grade sits on the right: $T_X B = B \otimes X$. A test on $A$ is a pair $(Y, E)$ of an ancilla $Y$ and an environment $E : A \otimes Y \to \Omega$ (which outputs a bit); two tests are equal when their ancillas coincide and their environments are $\cong_{\mathsf{M}}$-isomorphic. $\mathsf{Iso} := \mathsf{Std}(\mathsf{Mach}^{\mathrm{rev}}, \mathcal{E}^{\cong})$. (Agda: `Channel`, `Machine`, `TraceRel`, `_≅ᴹ_`, `machines`, `Test`, `SameTest`, `ℰ-tests`, `StdUC machines ℰ-tests`)

Because the equality is isomorphism, $f \leq_{\mathrm{UC}} g$ in Iso asserts *isomorphic* composites under every environment, which is stronger than indistinguishability; a failure of it is not an attack.

**Theorem 45** (Iso is grade-stable). Iso is grade-stable: a test with ancilla $Y$ on $B \otimes Z$ regroups into a test with ancilla $Z \otimes Y$ on $B$. Hence $\approx^{\mathrm{U}} = \approx_{\mathcal{E}}$ in Iso, and with the trivial ancilla $\approx_{\mathcal{E}}$ refines isomorphism under every environment. (Agda: `grade-stable`, `≈ᵁ⇔≈ℰ`, `≈ℰ⇒≅ℰ`)

**Proposition 46** (isomorphism is finer than bisimilarity). The identity machine with one redundant bit of state is not $\cong_{\mathsf{M}}$-isomorphic to the identity. (Agda: `padded-not-iso`)

## 3.7 Rel: relations

**Definition 47** (Rel). $\mathsf{Rel} := \mathsf{Std}((\mathsf{Rel}, \times, 1), \mathrm{Hom}(-, 1))$: sets and relations under the cartesian product; a test on $B$ is a subset of $B$, restricted along relations by preimage. A protocol $A \to X \times B$ relates inputs to (leakage, output) pairs. (Agda: `Rel`, `ℰ-tests`, `StdUC Rel ℰ-tests`)

**Theorem 48** (Rel is faithful). The test presheaf of Rel is faithful. Hence Rel is grade-stable, $\approx^{\mathrm{U}}$ is equality of relations, and $f \leq_{\mathrm{UC}} g$ iff $f = (s \times \mathrm{id}) \circ g$ for some relation $s$ (possibilistic simulation). (Agda: `ℰ-faithful`, `≈ᵁ⇒≈`, `≈C⇒≈ᵁ`, `≤UCᵈ⇔≤UC`)

**Theorem 49** (possibilistic one-time pad). Encryption with a uniform key is the relation "any ciphertext" after discarding the message; the one-time-pad channel emulates the ideal channel, emulation composes along relays, and leakage may always be added; the relay leaking the message does not emulate the ideal channel. (Agda: `otp-perfect`, `send≤ideal`, `relayed-otp`, `relayed-otp²`, `ideal≤leaky`, `send≤leaky`, `leaky-not-secure`)

**Theorem 50** (environment-blind robustness is incomplete). The leaky relay is outcome-included in the ideal channel, yet does not emulate it. Hence no level of environment-blind robust property preservation implies $\leq_{\mathrm{UC}}$ (Theorem 15(2)). (Agda: `relay-outcomes`, `blind-incomplete`)

Rel is the possibilistic one-shot model: a cheap sanity check of the abstract theory and the natural target of a cartesian (functions-under-$\times$) reading of Turing machines **DESIGN**. It is not a model of reactive UC, and no setup morphism connects it with the $+$-tensored reactive setups (Section 5.3).

## 3.8 Quantitative setups: QPM, QRS and the security-parameter tiers

The setups so far identify two tests or two protocols outright. Cryptographic statements carry an error, and the library keeps it in a quantitative variant of Definition 1.

**Definition 51** (quantitative UC setup). Fix an ordered error algebra $(\mathcal{R}, \oplus, \varepsilon_0, \sqsubseteq)$; the examples use $(\mathbb{Q}, +, 0, \leq)$ or functions $\mathbb{N} \to \mathbb{Q}$ pointwise. A *quantitative UC setup* $(\mathcal{C}, \mathcal{I}, \mathcal{M}, Q)$ is a UC setup whose test presheaf is replaced by a presheaf $Q$ of *approximation spaces*: each $Q(A)$ carries relations $e \approx_{\varepsilon} e'$ that hold reflexively at $\varepsilon_0$, are symmetric and

monotone in $\varepsilon$, and compose by $e \approx_\varepsilon e' \approx_\delta e'' \Rightarrow e \approx_{\varepsilon\oplus\delta} e''$; restriction along $h$ is nonexpansive. The *U-agreement at* $\varepsilon$ is

$$f \approx^{\mathrm{U}}_{\varepsilon} g :\Longleftrightarrow \forall W \in \mathcal{J}, e \in Q(T_{W\otimes X}B). \quad \mathrm{run}_W(f)(e) \approx_\varepsilon \mathrm{run}_W(g)(e),$$

and a *witness at* $\varepsilon$ is a simulator $s$ with $f \approx^{\mathrm{U}}_{\varepsilon} \mathrm{sub}_s \circ g$; write $\mathsf{Wit}_\varepsilon(f,g)$ if one exists. (Agda: `QUCSetup`, `_≈ᵁ[_]_`, `At`, `Witness`)

**Theorem 52** (quantitative metatheory). In every quantitative UC setup:

1. $\approx^{\mathrm{U}}_{\varepsilon}$ is reflexive at $\varepsilon_0$, symmetric, monotone, and $f \approx^{\mathrm{U}}_{\varepsilon} g \approx^{\mathrm{U}}_{\delta} h \Rightarrow f \approx^{\mathrm{U}}_{\varepsilon\oplus\delta} h$.
2. *Dummy adversary at a fixed error.* $\mathsf{Wit}_\varepsilon(f,g)$ iff for every adversary $a$ there is a simulator $s$ with $\mathrm{sub}_a \circ f \approx^{\mathrm{U}}_{\varepsilon} \mathrm{sub}_s \circ g$; the simulator is $a \circ s_0$, independent of $\varepsilon$.
3. *Composition.* $\mathsf{Wit}_\varepsilon(f,g)$ and $\mathsf{Wit}_\delta(g,h)$ give $\mathsf{Wit}_{\varepsilon\oplus\delta}(f,h)$; $\mathsf{Wit}_{\varepsilon_f}(f,g)$ and $\mathsf{Wit}_{\varepsilon_u}(u,v)$ give $\mathsf{Wit}_{\varepsilon_u\oplus\varepsilon_f}(u \bullet f, v \bullet g)$, with simulator $(\mathrm{id}\otimes t)\circ(s\otimes\mathrm{id})$ as in Theorem 8.
4. *Forgetting the error.* Identifying tests at $\varepsilon_0$ gives a UC setup $\mathbb{S}_0$ in which $\approx^{\mathrm{U}} \Longleftrightarrow \approx^{\mathrm{U}}_{\varepsilon_0}$ and $\leq_{\mathrm{UC}} \Longleftrightarrow \mathsf{Wit}_{\varepsilon_0}$. Identifying tests that are $\varepsilon$-close for every positive $\varepsilon$ gives a UC setup $\mathbb{S}_+$ in which $f \approx^{\mathrm{U}} g$ iff $f \approx^{\mathrm{U}}_{\varepsilon} g$ for every $\varepsilon > 0$, and $f \leq_{\mathrm{UC}} g$ iff *one* simulator witnesses every $\varepsilon > 0$ (a simulator depending on $\varepsilon$ is not enough).

(Agda: `ctx-refl`, `ctx-sym`, `ctx-trans`, `ctx-mono`, `dummy⇒universal`, `universal⇒dummy`, `at-trans`, `at-compose`, `underlying₀`, `underlying₊`, `zero-agreement`, `zero-emulation`, `positive-agreement`, `Witness₊`, `positive-emulation`, `Witness₊⇒Witness`)

Fixed positive-error closeness is not transitive at the same error, so it is not the hom-equality of any setup; only its all-positive closure in (4) is. This is why a quantitative statement is kept quantitative until the end and forgotten once.

### 3.8.1 QPM: the protocol model with errors kept

**Definition 53** (QPM). For verdict laws $d, d' \in D_p\ \mathsf{Bool}$, $d \approx_\varepsilon d'$ iff each verdict mass of either law is dominated by the other's up to $\varepsilon$ at every finite approximant (an *open* slack: the masses are lower reals). On tests, $e \approx_\varepsilon e'$ iff $\mathrm{Obs}(e \circ m) \approx_\varepsilon \mathrm{Obs}(e' \circ m)$ for every closure $m : 0 \to A$. Then $\mathsf{QPM} := (\mathbb{G}^o, \mathbb{G}^o, X \otimes (-), Q^o)$, with $Q^o(A) = \mathbb{G}^o(A, \Omega)$ under these relations. (Agda: `_≈ₚ[_]_`, `_≈ᵗ[_]_`, `Q`, `QSetup`, `QSetupᵒ`)

**Proposition 54** (QPM forgets to PM). (1) $f \approx^{\mathrm{U}}_{\varepsilon} g$ iff for every ancilla $Y$, test $t$ on $Y \otimes B$ and closure $m$ of $Y \otimes A$, the runs $\mathrm{Obs}(t \circ (\mathrm{id}\otimes f) \circ m)$ and $\mathrm{Obs}(t \circ (\mathrm{id}\otimes g) \circ m)$ are $\varepsilon$-close. (2) Two tests are equal in $\mathcal{E}^o$ iff they are $\varepsilon$-close for every $\varepsilon > 0$: $\mathcal{E}^o$ is the all-positive forgetting of $Q^o$, and $\mathsf{PM} = \mathsf{QPM}_+$ up to the order of the closure and error quantifiers. **INSTANCE** Hence, by Theorem 52(4) and Theorem 11 in both directions, $f \approx^{\mathrm{U}} g$ in PM iff $f \approx^{\mathrm{U}}_{\varepsilon} g$ in QPM for every $\varepsilon > 0$, and $f \leq_{\mathrm{UC}} g$ in PM iff one simulator is a QPM witness at every $\varepsilon > 0$. (Agda: `≈ℰ[]⇔≈ᵁ[]`, `Q₊⇔ℰ₊`, `≋⇔~₊`, `absorbᵁ`)

### 3.8.2 QRS: random systems with an exact error

The open slack of QPM does not descend to $\mathbb{G}^R$: the hom-equality $\approx_{\mathsf{R}}$ there is only the limit of $\Delta$ at zero (Proposition 24), and an open bound at $\varepsilon$ is not invariant under it. Its *closure* is: $d \bar{\approx}_\varepsilon d'$ iff $d \approx_{\varepsilon+\delta} d'$ for every $\delta > 0$.

**Definition 55** (QRS). $\mathsf{QRS} := (\mathbb{G}^R, \mathbb{G}^R, X \otimes (-), Q^R)$, with $Q^R(A) = \mathbb{G}^R(A, \Omega)$ and $e \approx_\varepsilon e'$ iff $\mathrm{Obs}(e \circ m) \bar{\approx}_\varepsilon \mathrm{Obs}(e' \circ m)$ for every closure $m$. (Agda: `_≈[_]_`, `closure`, `spaceᴿ`, `qevaluationᴿ`, `QSetupᴿ`)

**Theorem 56** (constructive cryptography at an exact $\varepsilon$). For PM protocols $f, g$, read in $\mathbb{G}^R$:

1. $f \approx^{\mathrm{U}}_{\varepsilon} g$ in QRS iff $f \approx^{\varepsilon}_{\mathsf{R}} g$: the U-agreement of QRS is the advantage bound, with no slack spent.
2. $g$ has a QRS witness at $\varepsilon$ for $f$, written $f \leq^{\varepsilon}_{\mathrm{UC}} g$, iff $\mathsf{Constructs}_\varepsilon(f,g)$ (Definition 27; [8] Def. 3 at $\varepsilon$).
3. *Along* rs: a QPM agreement (witness) at $\varepsilon$ gives a QRS agreement (witness) at $\varepsilon$; and $f \approx^{\mathrm{U}}_{\varepsilon} g$ in QRS iff $f \approx^{\mathrm{U}}_{\varepsilon+\delta} g$ in QPM for every $\delta > 0$, $f \leq^{\varepsilon}_{\mathrm{UC}} g$ in QRS iff one simulator is a QPM witness at every $\varepsilon + \delta$.
4. At $\varepsilon = 0$ it is emulation in RS: $\mathsf{rs}(f) \leq_{\mathrm{UC}} \mathsf{rs}(g)$ iff $f \leq^{0}_{\mathrm{UC}} g$.
5. ([8] Thm. 1 at a slack.) $f \leq^{0}_{\mathrm{UC}} f$; $f \leq^{\varepsilon}_{\mathrm{UC}} g \leq^{\varepsilon'}_{\mathrm{UC}} h \Rightarrow f \leq^{\varepsilon+\varepsilon'}_{\mathrm{UC}} h$; and $f \leq^{\varepsilon}_{\mathrm{UC}} g, u \leq^{\varepsilon'}_{\mathrm{UC}} v$ give $u \bullet f \leq^{\varepsilon'+\varepsilon}_{\mathrm{UC}} v \bullet g$.

(Agda: `≈ᵁᴿ[]⇔~ᴿ[]`, `_≤UCᴿ[]_`, `≤UCᴿ[]⇔Constructs`, `≈ᵁ[]⇒≈ᵁᴿ[]`, `Witness⇒≤UCᴿ[]`, `≈ᵁᴿ[]⇔≈ᵁ[]⁺`, `≤UCᴿ[]⇔Witness⁺`, `≤UCᴿ⇔≤UCᴿ[0]`, `≤UCᴿ[]-refl`, `≤UCᴿ[]-trans`, `≤UCᴿ[]-•`)

So the library's emulation order, kept quantitative over random systems, *is* Maurer and Renner's construction notion at every $\varepsilon$, and its composition theorem is the generic one of Theorem 52(3). PM's own $\approx^{\mathrm{U}}_{\varepsilon}$ matches it only up to every positive slack (item 3).

### 3.8.3 $\mathsf{PM}^{\omega}$ and $\mathsf{PM}^{\mathsf{N}}$: the vanishing and the negligible tier

**Definition 57** ($\mathsf{PM}^{\omega}$, $\mathsf{PM}^{\mathsf{N}}$). $\mathbb{G}^{o}_{\omega}$ is the category of $\mathbb{N}$-indexed families: objects are families $(A_n)_n$ of PM interfaces, morphisms are levelwise PM morphisms carrying a certificate of a resource rate polynomial in the security parameter, the tensor is levelwise. Families of laws are compared *eventually*: $\mu \approx^{\omega}_{\varepsilon} \nu$ iff $\mu_n \approx_{\varepsilon} \nu_n$ for all large $n$.

- $\mathsf{PM}^{\omega} := \mathsf{Std}(\mathbb{G}^{o}_{\omega}, \mathcal{E}^{\omega})$ identifies tests that are eventually $\varepsilon$-close for every $\varepsilon > 0$ under every closure: *vanishing advantage*.
- $\mathsf{PM}^{\mathsf{N}} := \mathsf{Std}\big(\mathbb{G}^{o}_{\omega}, \mathcal{E}^{\mathsf{N}}\big)$ identifies tests $t, t'$ when for every closure $m$ there is a negligible $\delta$ with $\mathrm{Obs}(t \circ m)_n \approx_{\delta(n)} \mathrm{Obs}(t' \circ m)_n$ for all $n$: *local negligibility*.

(Agda: `Obj^ω`, `_⇒^ω_`, `Fam^M`, `grading^o`, `_≈^ω[_]_`, `Evaluation^ω`, `ucSetup^ω`, `Evaluation^N`, `ucSetup^N`, `𝔾^o qevaluation^o grading^o`)

**Theorem 58** (the tiers).

1. *Concrete bounds.* Say $f \approx_{\varepsilon} g$ for $\varepsilon : \mathbb{N} \times \mathbb{N} \to \mathbb{Q}$ if at every ancilla context the level-$n$ advantage is at most $\varepsilon(n, q)$, $q$ the allowance the context affords. If $\varepsilon$ vanishes at every polynomial allowance, $f \approx^{\mathrm{U}} g$ in $\mathsf{PM}^{\omega}$ if it is negligible, so too, and the bound with its witness kept ($f \approx^{\mathsf{N}} g$) gives $f \approx^{\mathrm{U}} g$ in $\mathsf{PM}^{\omega}$ and $f \leq_{\mathrm{UC}} g$ in $\mathsf{PM}^{\mathsf{N}}$.
2. $\mathsf{PM}^{\omega}$ and $\mathsf{PM}^{\mathsf{N}}$ have the same category, grades and graded monad, on the nose: they differ only in their tests.
3. Each is the forgetting of one quantitative setup on $\mathbb{G}^{o}_{\omega}$: for $\mathsf{PM}^{\omega}$, $\approx^{\mathrm{U}}$ and $\leq_{\mathrm{UC}}$ coincide with those of the all-positive forgetting; for $\mathsf{PM}^{\mathsf{N}}$ only one direction holds — a bound at *one* negligible schedule good for every context is a $\mathsf{PM}^{\mathsf{N}}$ witness — and the converse (uniformisation) is open.

(Agda: `_≈ℰ[_]_`, `absorb`, `absorb-negl`, `_≈ℰ^n_`, `≈ℰ^n⇒≈^U`, `≈ℰ^n⇒≤UC`, `shared-computational`, `≈^U^ω⇔≈^U₊`, `≤UC^ω⇒≤UC₊`, `≤UC₊⇒≤UC^ω`, `uniform⇒local`, `≤UC^s⇒≤UC^N`, `uniform-witness⇒≤UC^N`)

Both tiers inherit Theorem 8, so the composition theorem of computational UC ([2]) holds in $\mathsf{PM}^{\mathsf{N}}$ with no further proof. The relation $\approx^{\mathrm{U}}$ of $\mathsf{PM}^{\omega}$ is too weak to carry a property whose error must stay negligible (a bad event of probability $\frac{1}{n+1}$ vanishes); $\mathsf{PM}^{\mathsf{N}}$ keeps the witness. A family tier over RS would now be possible — $\mathbb{G}^{R}$ has the quantitative readout of QRS — but needs a resource grading on $\mathbb{G}^{R}$, which is not built.

# 4 Designed setups (not formalised)

Nothing in this section is type-checked. Each setup is specified by a design document on the branch, and every statement is a **DESIGN** claim argued there.

## 4.1 $\mathsf{SSP}^{\mathsf{d}}$: distributional state-separating proofs

Proposition 40 shows that a deferred-sampling step is never a hom-equality of SSP. $\mathsf{SSP}^{\mathsf{d}}$ keeps SSP's objects, packages, composition and tests and coarsens $\approx_{\mathsf{p}}$ to a distributional equality of call trees, under which moving a uniform draw from the sender's step to the eavesdropper's poll is an equation. It is a quotient of SSP by a monoidal congruence below $\approx_{\mathcal{E}}$, so the identity-on-objects functor $\mathsf{SSP} \to \mathsf{SSP}^{\mathsf{d}}$ is a change of hom-equality with identity test map (Proposition 14) — provided the distributional equality is a monoidal congruence, which is the obligation. `examples-ssp-otp.md`

## 4.2 IPDL: interactive probabilistic dependency logic

Objects are polarised typed channel contexts pos $\rightleftharpoons$ neg under concatenation; a morphism $A \to B$ is an IPDL reaction network in flat form reading pos $A$, neg $B$ and defining neg $A$, pos $B$; composition is new $B$ in $(f \parallel g)$, identities are copy reactions, and the hom-equality is IPDL's exact equational theory ([14] §4.3). Tests are IPDL protocols into $\Omega$ modulo IPDL closures, read by lazy memoised evaluation into $D_p$ Bool. $\mathsf{IPDL} := \mathsf{Std}(\mathbb{I}, \mathcal{E}^{\mathsf{n}})$. The closest formal setup is PM: $\mathbb{I}$ is the G-construction of unpolarised nets. `uc-setup-ipdl.md`

## 4.3 CB: causal boxes

The classical fragment of [15] on positions $\mathbb{N}$: objects are pairs of slot alphabets under $\times$ (Portmann's parallel composition, Def. 6.1, is concurrent); a box runs each position in two phases, left-to-right then right-to-left, so its only delayed path is caller-input to caller-output; composition is Def. 6.6 in one pass, with no fixpoint; the hom-equality is behavioural equality under adaptive environments and the tests are Portmann's distinguishers (Def. 7.1). The paper proves no theorem of a traced symmetric monoidal category (its identity is not causal, §7); the two-phase discipline supplies one. `uc-setup-causal-boxes.md`

### 4.4 Sym and $\mathsf{Sym}_{\cong}$: the Dolev–Yao model

$\mathbb{DY}$ is the free monoidal category on *Dolev–Yao roles* — finite reactive programs whose outputs are recipes over their own frame (the roles of [17]) — with the free monoidal congruence as hom-equality. Tests are roles with an ancilla, as in Iso. In Sym two tests are equal when the same verdicts are possible under some globally fresh naming, run in PM at term alphabets; its kernel is Dolev–Yao trace (may-testing) equivalence ([16]), and Sym is grade-stable without hypotheses. $\mathsf{Sym}_{\cong}$ is the syntactic twin with the same $\mathbb{DY}$ and tests up to syntactic equality; all morphisms leave from it (Section 5.2). `uc-setup-symbolic.md`

### 4.5 $\mathsf{ND_A}$ and $\mathsf{ND_B}$: nondeterministic choice

The scheduler of a nondeterministic protocol is placed in the *grade*, as a negative factor.

**Definition 59** (choice factors, $\mathsf{ND_B}$, $\mathsf{ND_A}$). Let $\mathbb{M}$ be PM (or SSP). For a set $S$ the *choice factor* is the interface $C_S := (1, S)$: the protocol raises a request and receives a pick in $S$. A protocol with $n$ choice sites is $f : A \to \left(X \otimes C_{S_1} \otimes \cdots \otimes C_{S_n}\right) \otimes B$; a choose node of a call tree compiles to a request on its own factor and resumes with the pick.

- $\mathcal{G}_B$ is the full subcategory of grades $X \otimes C_{S_1} \otimes \cdots \otimes C_{S_n}$, $n \geq 0$, with their choice factors designated. It is closed under $\otimes$ up to the symmetry that moves choice factors to the right (present in the G-construction, not in the formalised $\mathbb{G}^o$), contains the unit at $n = 0$, and its inclusion into $\mathbb{M}$ is strong monoidal. Since $n = 0$ is allowed it contains every interface: $\mathsf{ND_B} := \mathsf{Std}(\mathbb{M})|_{\mathcal{G}_B}$ is $\mathsf{Std}(\mathbb{M})$ itself, with choice factors named.
- $\mathcal{G}_A \subseteq \mathcal{G}_B$ is the wide subcategory of morphisms $u \otimes \rho$, with $u$ arbitrary on $X$ and $\rho$ a *relabelling* of choice factors: a stateless wire that forwards each request and pulls the pick back along a function $S'_j \to S_i$ (so $\mathcal{G}_A \cong \mathcal{I} \times \mathsf{Alph}^{\mathrm{op}}$ on the choice part). $\mathsf{ND_A} := \mathsf{Std}(\mathbb{M})|_{\mathcal{G}_A}$.

Both are grade-side restrictions: the computations, the tests and $\mathcal{M}$ restricted along the inclusion (a graded monad pulls back along a lax monoidal functor) are those of $\mathsf{Std}(\mathbb{M})$, and

$$\mathsf{ND_A} \hookrightarrow \mathsf{ND_B} \hookrightarrow \mathsf{Std}(\mathbb{M})$$

are setup morphisms $(\mathrm{Id}, \mathsf{incl}, \mathrm{id}, \mathrm{id})$. (Agda: `pullback`)

**Who resolves.** In $\mathsf{ND_B}$ an adversary $a$ is any morphism of grades, so it may answer a real choice request from its own state and from what it learned on $X$, and a simulator may answer the ideal world's requests from what it learns from $a$: real choices are adversary-resolved, ideal ones simulator-resolved, which is Canetti's model, where the adversary controls delivery order ([3]; [25]). In $\mathsf{ND_A}$ no grade morphism can answer a request, so every request reaches the environment, which resolves the choices of both worlds, the simulator only renaming sites. This is the reading wanted for *specifications* ("send to $A$, $B$, $C$ in any order"): choose$[p, q]$ and choose$[q, p]$ emulate each other in $\mathsf{ND_A}$ with the swap of the two picks as simulator.

**Proposition 60** (emulation in $\mathsf{ND_A}$ implies emulation in $\mathsf{ND_B}$). DESIGN For protocols $f, g$ graded in $\mathcal{G}_A$: if $f \leq_{\mathrm{UC}} g$ in $\mathsf{ND_A}$ then $f \leq_{\mathrm{UC}} g$ in $\mathsf{ND_B}$. The converse fails.

*Proof idea.* $\mathcal{G}_A$ is wide in $\mathcal{G}_B$, so the two setups have the same grade objects, hence the same ancillas, prefix contexts and tests: $\approx^{\mathrm{U}}$ is literally the same relation in both. By dummy completeness (Theorem 8(2), valid in every setup), $f \leq_{\mathrm{UC}} g$ in $\mathsf{ND_A}$ gives $s_0 \in \mathcal{G}_A(Y, X)$ with $f \approx^{\mathrm{U}} \mathrm{sub}_{s_0} \circ g$; since $\mathcal{G}_A(Y, X) \subseteq \mathcal{G}_B(Y, X)$, the same $s_0$ is a dummy simulator in $\mathsf{ND_B}$. The converse fails because a $\mathsf{ND_B}$ simulator may answer requests, which no relabelling does. The inclusion preserves $\leq_{\mathrm{UC}}$ and does not reflect it; Theorem 10 is not needed (and its hypotheses fail: PM is not grade-stable, and incl is not full).

So a specification proved order-independent in $\mathsf{ND_A}$ is usable in the adversarially scheduled $\mathsf{ND_B}$. The previous design of `uc-setup-nondeterministic.md` — a choose constructor resolved only at observation, outcomes $\mathsf{Sched} \to \mathsf{Dist}_{\perp}\,\mathsf{Bool}$ under image equality — is superseded: its obstruction (a demonic closure sees a test's coins, the leak of [26]) does not arise, because a closed system has no choice factor left and is observed by the ordinary tests.

## 5 Morphisms between the setups

Solid arrows are formal setup morphisms, dashed arrows designed ones, red dotted arrows the candidate maps shown *not* to be morphisms (Section 5.3). Forgetting the error of a quantitative setup (QPM to PM, QRS to RS, the tiers to their quantitative families) is a change of kernel relation on one computational structure (Theorem 11), not drawn.

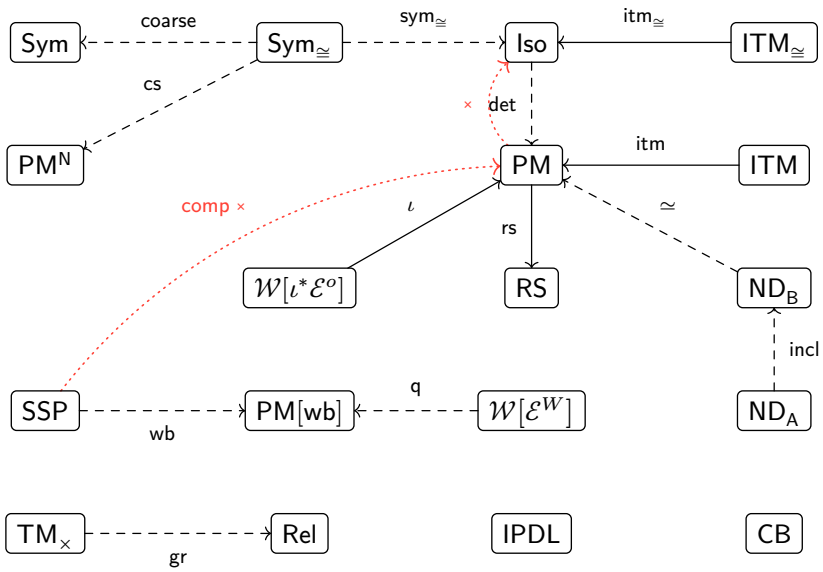



Figure 1: The morphism graph. IPDL and CB are islands; no morphism crosses between the $\times$-tensored setups ($\mathsf{TM}_\times$, Rel) and the $+$-tensored reactive ones.

## 5.1 Formal morphisms

### 5.1.1 rs : PM $\to$ RS

**Proposition 61** (the random-systems quotient). rs is the standard morphism (Proposition 13) of the quotient functor $\mathbb{G}^o \to \mathbb{G}^o/\approx_{\mathsf{R}}$ (Proposition 14(1)): identity on objects and on representatives, $\kappa = \mathrm{id}$, and test map the identity, Epi and Mono. It preserves and reflects emulation: $f \leq_{\mathrm{UC}} g$ in PM iff $\mathsf{rs}(f) \leq_{\mathrm{UC}} \mathsf{rs}(g)$ in RS (Theorem 28(1)); it reflects $\approx^{\mathrm{U}}$ ($f \approx_{\mathsf{R}} g \Rightarrow f \approx^{\mathrm{U}} g$), and with Theorem 17 the U-kernel of PM *is* its image's hom-equality. (Agda: `rewrite⇒RS`, `quotientF`, `descend^V-mono`, `≈^R⇒≈^U`, `≤UC-resp-≈^R`, `≤UC^R⇔Constructs₀`)

The preservation of $\leq_{\mathrm{UC}}$ is *not* Theorem 10(2) (RS's grade stability is not known); it is the explicit characterisation by constructions, which holds on both sides.

### 5.1.2 itm : ITM $\to$ PM

**Proposition 62** (interpreting Turing machines). $\mathsf{itm} = (F^p, F^p, \mathrm{id}, \nu^T)$ with $F^p$ the strict monoidal interpretation of Definition 41 and $\nu^T$ the identity on tests (the tests of ITM are PM's, pulled back), Epi and Mono. Hence $F^p$ preserves and reflects $\approx_{\mathcal{E}}$ (Theorem 43). Neither half of Theorem 10(2)–(3) applies: PM is not grade-stable and $F^p$ is not full. (Agda: `itm⇒prot`, `F^P`, `ν^T`, `ν^T-epi`, `ν^T-mono`, `≈E⇔`, `freeStrongMonoidal`)

### 5.1.3 itm$_{\cong}$ : ITM$_{\cong}$ $\to$ Iso

**Proposition 63** (Turing machines as relations). $\mathsf{itm}_{\cong} = (F^r, F^r, \mathrm{id}, \nu^r)$ with $F^r$ the halting-relation interpretation of Definition 42 and $\nu^r$ Epi and Mono. **INSTANCE** Since Iso is grade-stable (Theorem 45), Theorem 10(1)–(2) apply: $F^r$ preserves and reflects $\approx_{\mathcal{E}}$, and $f \leq_{\mathrm{UC}} g$ in ITM$_{\cong}$ implies $F^r f \leq_{\mathrm{UC}} F^r g$ in Iso, so an impossibility in Iso is one in ITM$_{\cong}$. (Agda: `itm⇒iso`, `F^R`, `ν^R`, `ν^R-epi`, `ν^R-mono`, `grade-stable`, `transfer`)

This is the only formal morphism whose transport of $\leq_{\mathrm{UC}}$ is the general theorem. The triangle ITM $\to$ PM, ITM$_{\cong}$ $\to$ Iso and det is expected to commute only on deterministic leaves, up to the mirror isomorphism of port trees **DESIGN**.

### 5.1.4 $\iota$ : $\mathcal{W}[\iota^*\mathcal{E}^o]$ $\to$ PM

The inclusion of well-bracketed machines, with the identity test map, Epi and Mono (Proposition 33). It preserves and reflects $\approx_{\mathcal{E}}$; like itm it transports nothing more, for the same two reasons.

### 5.1.5 The generic morphisms, instantiated

Change of kernel ck and the wide-subcategory inclusion incl (Section 1.4) have one concrete use each: the refinements between each tier and its quantitative family (Theorem 58(3), through Theorem 11) and $\mathsf{ND}_{\mathsf{A}} \to \mathsf{ND}_{\mathsf{B}}$ (Proposition 60, designed).

## 5.2 Designed morphisms

**Definition 64** (wb : SSP → PM[wb]). DESIGN On objects, flattening $A \mapsto$ flat $A$; on packages, compilation to a flat protocol (queries tagged with their channel, a reply on the wrong channel is an abort) followed by the machine semantics, which lands in well-bracketed machines (Theorem 30), then the quotient by $\sim_{\mathcal{W}}$. The test map sends a distinguisher to its compiled well-bracketed test. The two obligations are that the compiled identity wire is $\sim_{\mathcal{W}}$-equal to the identity and the compiled tensor $\sim_{\mathcal{W}}$-equal to PM's tensor — both fail at $\approx_{\mathsf{M}}$ (Section 5.3), and both are about well-bracketed environments only. (Agda: `toProtocol`, `toProtocol-resp`, `toProtocol-∘`, `morphism`, `morphism-resp`, `morphism-W`)

The test map is not Epi: a well-bracketed test can carry an irrational verdict mass, a distinguisher only a finite rational one. So the transport of the one-time pad (Theorem 39) to PM[wb] goes through the weaker hypothesis that $F$ preserves $\approx_{\mathcal{E}}$ (Agda: `Preserves-≈ℰ`), which has to be proved for wb directly.

**q : $\mathcal{W}[\mathcal{E}^{W}]$ → PM[wb].** The quotient morphism of Proposition 14(1) at $\sim_{\mathcal{W}}$. The general construction is formal; its instance does not type-check in reasonable time (two spellings of one universe level make the checker compare two quotient records by eta) DESIGN.

**det : Iso → PM.** DESIGN Three model changes are needed first: Iso machines get an initial state and $\cong_{\mathsf{M}}$ must preserve it; determinism (at most one successor) is imposed as a wide subcategory, which is closed under composition, tensor and the graded-monad structure; and silence is removed. Then $F$ runs a machine in $D_p$ from its initial state, and $\nu(Y, E) = FE \circ (\mathrm{id} \otimes \bot_Y) \circ \rho^{-1}$ closes the ancilla with the divergent process. Without the changes only the stateless deterministic fragment maps.

**sym$_{\cong}$ : Sym$_{\cong}$ → Iso, coarse : Sym$_{\cong}$ → Sym, cs : Sym$_{\cong}$ → PM$^{\mathsf{N}}$.** DESIGN sym$_{\cong}$ is the free strong monoidal functor on roles into (reversed) relational machines. coarse is a change of kernel from syntactic equality to Dolev–Yao trace equivalence. cs is a standard morphism into the negligible tier, unconditional; *computational soundness* is then a separate theorem over it, under the usual side conditions (IND-CPA with INT-CTXT, or IND-CCA2 with EUF-CMA; typing and tagging) — the test map being Mono gives completeness, not soundness. Nothing leaves Sym itself.

**gr : TM$_{\times}$ → Rel.** DESIGN The cartesian reading of Turing machines (bitstring tuples under $\times$, computable functions) maps strongly into Rel: a function to its graph, a computable predicate to its subset.

## 5.3 Non-morphisms

**Proposition 65** (obstructions). None of the following is a setup morphism.

1. SSP → PM *by compilation.* The compiled identity wire is not $\approx_{\mathsf{M}}$-equal to the identity of PM: the copycat accepts a reply at any time, while the compiled wire demands a query first, so in the idle state one diverges and the other answers. Moreover PM's tensor interleaves, which SSP's cannot. Both are observable through an ancilla, and both counterexamples are non-well-bracketed contexts. Compilation preserves hom-equality and composition: it is a semifunctor comp. DESIGN (argued in the source of `morphism`, not stated as a theorem)
2. PM → Iso. A functor must send $\approx_{\mathsf{M}}$ to $\cong_{\mathsf{M}}$, but simulation-equivalent machines with different state sets have no state bijection: the identity padded by a redundant bit is not $\cong_{\mathsf{M}}$-isomorphic to the identity (Proposition 46), while it is $\approx_{\mathsf{M}}$-equal to it in PM.
3. *Between $\times$- and $+$-tensored setups.* $\kappa$ would be a reactive machine receiving one pair and emitting two messages, or the converse, while a reactive step emits at most one message per input: the crossing is lax, and Definition 9 allows only oplax. DESIGN
4. IPDL → PM, IPDL → RS. The exchange and unused-channel laws of IPDL hold only against IPDL-shaped environments; a PM closure distinguishes them with advantage 1, so the compile fails both $F(\mathrm{id}) = \mathrm{id}$ and the respect of hom-equality. A semifunctor survives: attacks transfer from IPDL to PM, security does not. DESIGN
5. CB *to or from any other setup.* An internal round of the G-construction costs at least one causal position, so $F(g \circ f) = Fg \circ Ff$ fails; Portmann's synchronous tensor has no image under interleaving. Causal boxes are reachable only laxly. DESIGN
6. Sym → RS, Sym → Iso. Names are guessed with positive probability, so exact soundness fails; only the syntactic twin Sym$_{\cong}$ maps. DESIGN
7. *A change of kernel between $\mathcal{W}[\iota^{*}\mathcal{E}^{o}]$ and $\mathcal{W}[\mathcal{E}^{W}]$, in either direction* (Section 3.3). DESIGN
8. *A morphism reflecting emulation from* ND$_{\mathsf{B}}$ *to* ND$_{\mathsf{A}}$. The inclusion preserves $\leq_{\mathrm{UC}}$ and does not reflect it (Proposition 60). DESIGN

(Agda: `morphism`, `padded-not-iso`)

# 6 Open problems

1. *The call/return bridge.* Instantiate $\mathsf{q}$; prove the two obligations of $\mathsf{wb}$; prove that $\mathsf{wb}$ preserves $\approx_{\mathcal{E}}$, and transport the one-time pad to $\mathsf{PM}[\mathsf{wb}]$.
2. *Non-adaptive equivalence.* $\approx_{\mathsf{B}} \Rightarrow \approx_{\mathsf{R}}$ beyond finite alphabets (Proposition 22; [9] Thm. 2), or a reason to keep $\approx_{\mathsf{R}}$.
3. *Grade stability* of $\mathsf{SSP}$, $\mathsf{RS}$ and $\mathsf{PM}[\mathsf{wb}]$: neither proved nor refuted.
4. *Reactive simulatability* and the trace-property characterisation of [4] §8.2.1 (Section 1.6).
5. *Uniformisation* in the negligible tier: local negligibility implies a uniform negligible witness? (Theorem 58(3)).
6. *Constructive cryptography beyond Thm. 1:* availability, parallel composition (needs a braiding on $\mathbb{G}^{o}$), amplification (needs conditional probabilities), and a family tier over $\mathsf{RS}$ (needs a resource grading on $\mathbb{G}^{R}$).
7. *Feasibility.* Polynomial time is not a property of morphisms ([24] §1.5); computable tests of their own for $\mathsf{ITM}$ would need a density theorem.
8. $\mathsf{Iso}$ *into* $\mathsf{PM}$*:* the three model changes behind $\mathsf{det}$.
9. *The designed setups* of Section 4, starting with $\mathsf{ND}_{\mathsf{A}}$ (choice factors, relabellings, Proposition 60) and $\mathsf{SSP}^{\mathsf{d}}$ (the distributional congruence).
10. *Computational soundness* over $\mathsf{cs}$: the side-conditioned theorem and its transport, which needs a preservation hypothesis in place of Epi and grade stability.

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